

Lorentz Complex Rigidity, Hodge Constitutive Response, and Quantum Source Transduction in Motion–Staticity Geometry

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September 7, 2026

Abstract We investigate whether the complex structure associated with quantum comparison in the motion–staticity (MS) reconstruction can be related, without literal identification, to the Hodge structure and constitutive response of Lorentzian gravity.

The analysis begins with a dimension-independent one-time Lorentz family. For the gravitational generator carrier $\Lambda^2 V \simeq \mathfrak{so}(1, d-1)$, we classify Lorentz-equivariant real endomorphisms and ask when the commutant contains a real complex structure $J^2 = -I$. Within the adjoint/bivector representation class considered here, this requirement selects $d = 4$, where

$$\text{End}_{SO^+(1,3)}(\Lambda^2 V) = \mathbb{R}I \oplus \mathbb{R}\star$$

and the only invariant real complex structures are $J = \pm\star$. If the projective quantum-comparison sector and the reconstructed Lorentz geometry realize the same $\text{Spin}^+(1, 3) \simeq SL(2, \mathbb{C})$ cover, the real-linear complex structure $X \mapsto iX$ on $\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$ is transported precisely to $\pm\star$ on Lorentz bivectors.

An independent boundary argument reaches the same spacetime dimension. If the null-direction sphere S^{d-2} supports a fundamental rank-one complex helicity bundle with nonzero first Chern class, then $H^2(S^{d-2}; \mathbb{Z}) \neq 0$, forcing $d = 4$.

The same bivector commutant constrains a local real Lorentz-equivariant gravitational constitutive map to

$$C = \alpha I + \beta \star = K e^{\theta \star}.$$

A further comparison–constitutive compatibility condition, formulated as invariance of the constitutive graph under a doubled quarter-turn complex structure, selects conditionally the pure Hodge direction $C \propto \star$. The magnitude K_G is interpreted as gravitational stiffness, with K_G^{-1} controlling source-to-geometry compliance.

For an algebraic spin source, $T = L_e \tau$, promotion of τ to a quantum operator induces $\widehat{T} = L_e \widehat{\tau}$. Expectation values, fixed-order connected correlators, commutators, and appropriately defined generating functionals are therefore transferred through the classical response map. This defines source-induced operator-valued geometry but does not by itself constitute intrinsic quantum gravity.

Causal response is described using retarded Kubo kernels and quantum noise. In equilibrium the corresponding fluctuation–dissipation relation links noise and spectral response, while the reduced first-order symplectic normalization gives the quantum conjugacy scale

$$\hbar K_G^{-1} \propto \frac{\hbar G}{c^3} = \ell_P^2.$$

This identifies the Planck area as a quantum-response scale, not as a minimum-area theorem.

Finally, we distinguish local representation-theoretic rigidity from infrared dynamical closure. A genuine pure-Hodge directional limit is formulated using the normalized operator $\widehat{C}_\mu = C_\mu/K(\mu)$ and the condition $\widehat{C}_\mu^2 + I \rightarrow 0$, together with a finite positive infrared stiffness $K(\mu) \rightarrow K_G > 0$. Mixing with additional comparison sectors is constrained by symmetry and, on positive static sectors, by Schur-complement stability bounds. The resulting framework therefore specifies a mathematically sharp candidate architecture for an infrared Einstein response while leaving its microscopic dynamical realization open.

Keywords: motion–staticity geometry; Lorentz representation theory; Hodge duality; spin cover; constitutive gravity; dimensional rigidity; helicity line bundles; quantum source transduction; gravitational response; Kubo susceptibility; quantum noise; infrared Einstein closure.

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1 Introduction

A recurring difficulty in attempts to reconstruct quantum geometry is that several mathematically similar structures appear at different conceptual levels and are easily identified too early. Complex multiplication occurs in quantum state spaces; Hodge duality acts on differential forms and Lorentz bivectors; first-order gravitational theories contain constitutive maps between geometric field-strength and excitation variables; quantum matter generates operator-valued sources; and effective gravitational dynamics is characterized by response kernels, fluctuations, and running couplings. The fact that several of these structures involve operators squaring to $-I$ does not by itself make them the same object.

The purpose of the present paper is to determine which relations among these structures follow from representation theory and response theory, which require additional compatibility assumptions, and which remain genuinely dynamical.

The analysis is developed within the motion–staticity (MS) reconstruction program. The broader aim of that program is to treat spacetime geometry, projective quantum comparison, and internal relational structure as logically distinct sectors and to investigate how they may arise from a common operational comparison framework. The present paper addresses the gravitational-response branch of this problem.

The central question is the following. Suppose coherent quantum comparison carries a nontrivial complex structure. Under what conditions can that complex structure be represented geometrically on the Lorentz generator sector reconstructed from operational comparison? If such a representation exists, does it constrain spacetime dimension, the form of the gravitational constitutive map, or the response of geometry to quantum matter?

The first step is deliberately dimension independent. Instead of assuming three spatial dimensions from the outset, we write the generalized MS state relation as

$$D \in \mathbb{R}^m, \quad |D|^2 + S^2 = 1, \quad S \geq 0, \quad (1)$$

and associate the operational Lorentz reconstruction with spacetime dimension

$$d = m + 1. \quad (2)$$

The gravitational Lorentz-generator carrier is then

$$W_d = \Lambda^2 V \simeq \mathfrak{so}(1, d - 1). \quad (3)$$

We ask whether W_d admits a real Lorentz-equivariant complex structure

$$J_G : W_d \longrightarrow W_d, \quad J_G^2 = -I. \quad (4)$$

This is a representation-theoretic question about the commutant

$$\text{End}_{SO^+(1,d-1)}(W_d).$$

The answer yields the first rigidity result of the paper. Within the one-time Lorentz adjoint/bivector family analyzed below, an invariant real complex structure of this type occurs in four spacetime dimensions. In that case

$$\text{End}_{SO^+(1,3)}(\Lambda^2 V) = \mathbb{R}I \oplus \mathbb{R}\star, \quad (5)$$

and Lorentzian Hodge duality satisfies

$$\star^2 = -I \quad (6)$$

on two-forms. Consequently,

$$J_G = \pm \star \quad (7)$$

are the only Lorentz-equivariant real complex structures on this carrier.

This result should not be paraphrased as a literal identification

$$i = \star.$$

Scalar multiplication by i on a quantum state space, multiplication by i on the complex Lorentz spin-cover algebra, and Hodge duality on a real bivector space are different operations.

The relevant bridge appears at the level of the Lorentz spin cover. Regarded as a real Lie algebra,

$$\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$$

possesses the canonical real-linear complex structure

$$J_{\text{spin}} X = iX. \quad (8)$$

The differential of the Lorentz covering map gives

$$d\pi : \mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}} \xrightarrow{\cong} \mathfrak{so}(1, 3) \simeq \Lambda^2 V. \quad (9)$$

Transporting J_{spin} through this isomorphism produces a Lorentz-equivariant real complex structure on the bivector carrier, and the commutant classification then forces

$$d\pi J_{\text{spin}} d\pi^{-1} = \pm \star. \quad (10)$$

The mathematical statement Eq. (10) is exact once the spin-cover identification has been fixed. Its physical application to MS comparison is conditional. The nontrivial projective quantum-comparison sector does not by itself prove that the quantum carrier and the operationally reconstructed Lorentz geometry use the same local $\text{Spin}^+(1, 3)$ representation. We therefore keep the common-spin-cover requirement explicit rather than treating it as a consequence of projective quantum structure alone.

A second and independent dimensional test arises at the projective null boundary. In d spacetime dimensions the sphere of null directions is

$$S^{d-2}. \quad (11)$$

Suppose that a fundamental finite-helicity boundary sector is represented by a complex line bundle

$$L \longrightarrow S^{d-2} \quad (12)$$

with

$$c_1(L) \neq 0. \quad (13)$$

Since

$$c_1(L) \in H^2(S^{d-2}; \mathbb{Z})$$

and

$$H^2(S^n; \mathbb{Z}) = 0 \quad \text{for } n \neq 2,$$

this rank-one Chern mechanism independently selects

$$d = 4. \quad (14)$$

The bulk and boundary arguments are logically independent. The first uses the real commutant of the Lorentz adjoint representation. The second uses the integral cohomology of the null-direction sphere. Their common four-dimensional conclusion therefore constitutes a conditional bulk–boundary double rigidity rather than two reformulations of the same proof.

The boundary statement has a restricted scope. Higher-dimensional massless theories can possess non-Abelian little-group representations and higher-rank polarization bundles. The Chern argument therefore does not exclude massless quantum theories in dimensions other than four. It selects four dimensions only under the stated rank-one complex line-bundle assumption.

Once four-dimensional Lorentz bivectors have been selected, the same commutant controls the local algebraic gravitational constitutive response. Let B denote the bivector-type geometric field-strength variable and H the corresponding excitation. A local real-linear Lorentz-equivariant constitutive law

$$H = C(B) \quad (15)$$

must have

$$C = \alpha I + \beta \star. \quad (16)$$

For $C \neq 0$, this can be written

$$C = K e^{\theta \star}, \quad K = \sqrt{\alpha^2 + \beta^2} > 0. \quad (17)$$

This decomposition separates two physically different quantities. The magnitude K measures the overall response stiffness, while the angle θ specifies the constitutive orientation inside the invariant two-dimensional response algebra.

Lorentz covariance alone does not select

$$\theta = \pm \frac{\pi}{2}.$$

To obtain the pure Hodge constitutive direction we introduce an additional comparison–constitutive compatibility condition.

On the doubled constitutive response space

$$W \oplus W$$

define

$$J_\zeta(B, H) = (\zeta H, -\zeta^{-1} B), \quad J_\zeta^2 = -I. \quad (18)$$

The constitutive law defines the graph

$$\Gamma_C = \{(B, CB)\}.$$

Requiring

$$J_\zeta \Gamma_C = \Gamma_C \quad (19)$$

is equivalent to

$$C^2 = -\zeta^{-2} I. \quad (20)$$

Combining this with Eq. (16) gives

$$C = \pm \zeta^{-1} \star. \quad (21)$$

Thus complex graph invariance removes the identity channel and selects the pure Hodge ray.

This is a conditional structural result, not a consequence of Lorentz covariance alone. In particular, the doubled space (B, H) is a constitutive input–output space and should not automatically be identified with the canonical gravitational phase space. A microscopic theory would have to derive why the complex structure of quantum comparison is represented on gravitational response variables by Eq. (18).

The constitutive magnitude has a different logical status. Writing the pure-Hodge branch as

$$C = \varepsilon_C K_G \star, \quad \varepsilon_C = \pm 1, \quad K_G > 0, \quad (22)$$

the quantity K_G is not fixed by representation theory. It instead has a natural response-theoretic interpretation.

For an algebraic matter spin source τ , the connection equation may be written schematically as

$$C \mathcal{T}_e(T) = \tau. \quad (23)$$

On an invertible response channel,

$$T = L_e \tau, \quad L_e = \mathcal{T}_e^{-1} C^{-1}. \quad (24)$$

In the pure-Hodge branch,

$$L_e \propto K_G^{-1}.$$

The inverse constitutive stiffness therefore measures the compliance of the torsional geometric response to a normalized spin source.

This interpretation extends to dynamical channels. If a reduced infrared inverse response kernel factorizes as

$$\Gamma_R^{(2)} = K_G D_R, \quad (25)$$

then

$$K_R = K_G^{-1} D_R^{-1}. \quad (26)$$

With Einstein normalization,

$$K_G \propto \frac{c^3}{G}, \quad (27)$$

so Newton's coupling may be interpreted as proportional to gravitational compliance.

The statement is operational but not predictive of the numerical value of G . Causal response and spectral information constrain the analytic and dissipative structure of the susceptibility, while

local counterterms, subtraction constants, renormalization conditions, and ultraviolet physics remain capable of shifting the real local coefficient.

Quantum matter introduces a further layer. Replacing the classical source by an operator gives

$$\widehat{T} = L_e \widehat{\tau}. \quad (28)$$

This immediately transfers quantum statistics from matter to geometry.

For any fixed multilinear ordering prescription O ,

$$\begin{aligned} \langle O \widehat{T}(x_1) \cdots \widehat{T}(x_n) \rangle_c \\ = \left(\bigotimes_{r=1}^n L_{x_r} \right) \langle O \widehat{\tau}(x_1) \cdots \widehat{\tau}(x_n) \rangle_c. \end{aligned} \quad (29)$$

Similarly,

$$[\widehat{T}(x), \widehat{T}(y)] = L_x[\widehat{\tau}(x), \widehat{\tau}(y)] L_y^T. \quad (30)$$

A state may consequently satisfy

$$\langle \widehat{T} \rangle = 0$$

while retaining nonzero torsional variance and higher connected correlations.

This phenomenon will be referred to as *source-induced operator geometry*. It is quantum in the precise sense that geometric observables inherit noncommutativity and quantum correlations from their source. It does not, however, imply an independent quantization of the gravitational field.

This distinction is central:

$$\text{operator-valued geometric response} \not\Rightarrow \text{intrinsic quantum gravity}. \quad (31)$$

The same qualification applies to generating functionals. The ordinary characteristic functional

$$\left\langle \exp \left[\frac{i}{\hbar} \widehat{\tau}(J) \right] \right\rangle$$

generates the corresponding Weyl/symmetrized hierarchy. Time-ordered correlators require a time-ordered exponential, while more general real-time operator orderings require the corresponding multi-source or Schwinger–Keldysh construction. The linear source-transduction theorem itself remains valid once the desired ordering prescription has been fixed.

To characterize causal quantum response, we introduce the retarded Kubo kernel

$$K_{AB}^R(x, y) = -\frac{i}{\hbar} \theta(x^0 - y^0) \langle [\widehat{O}_A(x), \widehat{O}_B(y)] \rangle, \quad (32)$$

together with the symmetrized noise kernel

$$N_{AB}(x, y) = \frac{1}{2} \langle \{ \Delta \widehat{O}_A(x), \Delta \widehat{O}_B(y) \} \rangle. \quad (33)$$

In a KMS equilibrium state these quantities satisfy a fluctuation–dissipation relation. A Fourier representation in frequency requires stationarity; a simultaneous diagonal momentum-space representation $K(\omega, \mathbf{k})$ additionally requires spatial translation invariance. This distinction is important when the response is considered on nonuniform backgrounds.

The first-order symplectic normalization gives a complementary meaning to the same gravitational stiffness. On an appropriate reduced canonical channel,

$$\Omega_{\text{grav}} \sim K_G \int_{\Sigma} \delta(\star B) \wedge_F \delta\omega, \quad (34)$$

so the reduced bracket scales as

$$\{\omega, \star B\}_{\text{red}} \sim K_G^{-1}. \quad (35)$$

After quantization,

$$[\widehat{\omega}, \widehat{\star B}]_{\text{red}} \sim i\hbar K_G^{-1}. \quad (36)$$

Using Eq. (27), one obtains the characteristic area scale

$$\hbar K_G^{-1} \propto \frac{\hbar G}{c^3} = \ell_P^2. \quad (37)$$

This gives the Planck area a response-theoretic interpretation:

$$\ell_P^2 \sim \hbar \times \text{gravitational compliance}.$$

No minimum-area spectrum follows from this scaling relation alone.

Finally, the paper distinguishes exact local rigidity from infrared effective closure. At a coarse-graining scale μ , the projected local constitutive channel may take the form

$$C_\mu = K(\mu) e^{\theta(\mu)\star}. \quad (38)$$

Representation theory determines the possible local algebraic directions, but it does not imply that quantum coarse graining remains confined to this two-dimensional response plane or preserves the pure-Hodge ray.

Higher derivatives, nonlocal form factors, torsional operators, additional fields, and cross-sector mixing may all be generated.

Moreover, convergence of an unnormalized constitutive defect is not sufficient to establish convergence of the response direction if the overall magnitude tends to zero. For

$$K(\mu) > 0,$$

the appropriate directional variable is

$$\widehat{C}_\mu = \frac{C_\mu}{K(\mu)}. \quad (39)$$

Pure-Hodge directional closure is characterized by

$$\widehat{C}_\mu^2 + I \longrightarrow 0, \quad (40)$$

while a complete Einstein infrared target additionally requires

$$K(\mu) \longrightarrow K_G > 0. \quad (41)$$

Coupling to other comparison sectors introduces further constraints. On a positive static finite-mode sector, a quadratic stiffness matrix

$$\begin{pmatrix} K_G & M \\ M^\dagger & K_I \end{pmatrix}$$

produces the effective gravitational Schur complement

$$K_G^{\text{eff}} = K_G - MK_I^{-1}M^\dagger. \quad (42)$$

Under the appropriate finite-dimensional or bounded-positive-operator hypotheses,

$$0 < K_G^{\text{eff}} \leq K_G.$$

For one mode, stability requires

$$|m|^2 < K_G K_I. \quad (43)$$

Thus a comparison sector becoming soft in the infrared cannot remain arbitrarily strongly mixed with a stable finite-stiffness gravitational static mode.

Symmetry independently constrains the existence of such mixing. A direct bilinear coupling between gravitational and internal comparison modes requires a nonzero invariant intertwiner between their representations. The mere coexistence of a Lorentz connection and an internal gauge connection is insufficient to imply direct mixing or unification.

The results of the paper therefore fall into three logical levels.

At the first level are mathematical consequences of stated representation and response assumptions: the Lorentz bivector commutant, the four-dimensional invariant-complex-structure result, the helicity–Chern dimensional theorem, the constitutive graph criterion, and the linear source-transduction identities.

At the second level are conditional structural bridges: the transfer of the spin-cover complex structure to Hodge duality, the representation of comparison complexity on the doubled constitutive response, and the resulting pure-Hodge constitutive selection.

At the third level are dynamical questions: why the same Lorentz spin cover is selected, why the doubled complex response structure is realized, whether intrinsic gravitational quantum degrees of freedom emerge, what fixes the finite renormalized value of K_G , and whether the infrared response actually approaches an Einstein regime.

The structure of the paper follows these distinctions.

Section 2 classifies Lorentz-equivariant complex structures on the bivector/adjoint carrier and establishes the four-dimensional bulk rigidity. Section 3 transports the canonical complex structure of the Lorentz spin-cover algebra to Hodge duality and develops the independent null-boundary Chern argument. Section 4 classifies local constitutive response and formulates the conditional pure-Hodge graph-invariance criterion. Section 5 develops quantum source transduction, connected correlators, commutators, comparison-loop readout, and the distinction between source-induced and intrinsic quantum geometry. Section 6 interprets gravitational stiffness through Kubo response, noise, fluctuation–dissipation relations, and the reduced symplectic normalization. Section 7 studies constitutive renormalization flow, normalized infrared Hodge closure, comparison-sector mixing, Schur complements, and symmetry constraints. Section 8 separates the resulting mathematical statements, conditional physical identifications, limitations, and open problems. Section 9 summarizes the resulting gravitational-response architecture.

The objective is therefore not to claim that Einstein gravity or intrinsic quantum gravity has already been derived from MS kinematics. The more limited and mathematically sharper claim is that the comparison framework identifies a distinctive four-dimensional Lorentz complex structure, a corresponding pure-Hodge candidate constitutive direction, and a precise response-theoretic route by which quantum matter can generate correlated operator-valued geometry. A microscopic theory must still explain why these structures are selected dynamically and why their effective long-distance response approaches a stable finite-stiffness Einstein regime.

2 Lorentz-Equivariant Complex Structures and Four-Dimensional Rigidity

The complex structure appearing in quantum theory is usually introduced at the level of a complex Hilbert space. In the present reconstruction, however, the question of interest is more restrictive. We ask whether the gravitational comparison sector itself admits an intrinsic real-linear endomorphism that plays the role of multiplication by the complex unit and is compatible with the observer symmetry.

To avoid building the answer into the dimensional assumptions, we first generalize the local MS kinematics. Let the motion variable have m spatial components,

$$D \in \mathbb{R}^m, \quad |D|^2 + S^2 = 1, \quad S > 0, \quad (44)$$

and write

$$d = m + 1 \quad (45)$$

for the corresponding spacetime dimension. We assume that the operational reconstruction extends to this generalized setting and produces a d -dimensional Lorentz vector space (V, η) of signature $(-, +, \dots, +)$, with connected observer group

$$G_d := SO^+(1, d-1). \quad (46)$$

The arguments below require only this Lorentz representation-theoretic output; they do not depend on a detailed dynamical realization of the generalized compact state.

The natural carrier of infinitesimal Lorentz comparison is the bivector space

$$W_d := \Lambda^2 V. \quad (47)$$

The Lorentz metric gives a canonical equivariant identification between bivectors and infinitesimal Lorentz generators. Explicitly, define

$$\iota_\eta : \Lambda^2 V \longrightarrow \mathfrak{so}(1, d-1) \quad (48)$$

by

$$\iota_\eta(u \wedge v)(w) = \eta(v, w)u - \eta(u, w)v. \quad (49)$$

This is an isomorphism of real vector spaces. Moreover,

$$\iota_\eta(g \cdot B) = \text{Ad}_g(\iota_\eta(B)) \quad (50)$$

for every $g \in G_d$ and $B \in \Lambda^2 V$. Accordingly,

$$W_d \simeq \mathfrak{g}_d, \quad \mathfrak{g}_d := \mathfrak{so}(1, d-1), \quad (51)$$

as real G_d -modules.

The gravitational complex-structure problem can therefore be formulated without introducing spinors.

Definition 2.1 (Lorentz-equivariant gravitational complex structure). *A Lorentz-equivariant gravitational complex structure in spacetime dimension d is a real-linear map*

$$J_G : W_d \longrightarrow W_d \quad (52)$$

such that

$$J_G^2 = -I \quad (53)$$

and

$$J_G(g \cdot B) = g \cdot J_G(B) \quad \text{for all } g \in G_d, B \in W_d. \quad (54)$$

This requirement is stronger than ordinary complexification. Every real representation admits the formal extension

$$W_d^{\mathbb{C}} := W_d \otimes_{\mathbb{R}} \mathbb{C},$$

but such a construction introduces the complex scalar externally. Definition 2.1 instead requires the real gravitational carrier itself to possess an observer-invariant endomorphism whose square is $-I$.

The problem is therefore controlled by the real commutant

$$C_d := \text{End}_{G_d}(W_d). \quad (55)$$

Since $G_d = SO^+(1, d-1)$ is connected, commuting with the group action is equivalent to commuting with the differentiated Lie-algebra action. Using the equivariant identification $W_d \simeq \mathfrak{g}_d$, one obtains

$$C_d \simeq \text{End}_{\text{ad } \mathfrak{g}_d}(\mathfrak{g}_d) := \{T \in \text{End}_{\mathbb{R}}(\mathfrak{g}_d) ; T \text{ ad}_X = \text{ad}_X T \text{ for all } X \in \mathfrak{g}_d\}. \quad (56)$$

Thus the dimensional problem is governed by the real centroid of the Lorentz Lie algebra.

The exceptional case is $d = 4$.

Theorem 2.2 (Lorentz bivector commutant). *Let $d \geq 2$, and let*

$$W_d = \Lambda^2 \mathbb{R}^{1, d-1}$$

carry the natural action of $G_d = SO^+(1, d-1)$. Then:

1. *for $d = 2$,*

$$\text{End}_{G_2}(W_2) = \mathbb{R}I; \quad (57)$$

2. *for $d = 3$ and for every $d \geq 5$,*

$$\text{End}_{G_d}(W_d) = \mathbb{R}I; \quad (58)$$

3. *for $d = 4$,*

$$\text{End}_{G_4}(W_4) = \mathbb{R}I \oplus \mathbb{R}\star, \quad (59)$$

where, after choosing the spacetime orientation, $\star : \Lambda^2 V \rightarrow \Lambda^2 V$ is the Lorentzian Hodge operator, satisfying

$$\star^2 = -I \quad (60)$$

on bivectors.

Proof. For $d = 2$, the bivector space is one-dimensional:

$$\dim W_2 = 1.$$

Hence every real endomorphism of W_2 is a real scalar multiple of the identity, proving Eq. (57).

Now let $d = 3$ or $d \geq 5$. By Eq. (51), it is sufficient to determine the commutant of the adjoint representation of $\mathfrak{g}_d = \mathfrak{so}(1, d-1)$. Its complexification is

$$\mathfrak{g}_d^{\mathbb{C}} := \mathfrak{g}_d \otimes_{\mathbb{R}} \mathbb{C} \simeq \mathfrak{so}(d, \mathbb{C}). \quad (61)$$

For $d = 3$ and $d \geq 5$, $\mathfrak{so}(d, \mathbb{C})$ is a complex simple Lie algebra.

Let

$$T \in \text{End}_{G_d}(W_d).$$

After identifying $W_d \simeq \mathfrak{g}_d$, complexify T to

$$T_{\mathbb{C}} : \mathfrak{g}_d^{\mathbb{C}} \longrightarrow \mathfrak{g}_d^{\mathbb{C}}.$$

The map $T_{\mathbb{C}}$ commutes with ad_X for every $X \in \mathfrak{g}_d^{\mathbb{C}}$. Since the adjoint representation of a complex simple Lie algebra is irreducible, Schur's lemma gives

$$T_{\mathbb{C}} = \lambda I \quad (62)$$

for some $\lambda \in \mathbb{C}$.

Because $T_{\mathbb{C}}$ is the complexification of a real map, it commutes with the conjugation whose fixed-point set is $\mathfrak{g}_d$. Hence $\lambda = \bar{\lambda}$, so $\lambda \in \mathbb{R}$. This proves Eq. (58).

It remains to consider $d = 4$. On the oriented Lorentz vector space V , the Hodge operator

$$\star : \Lambda^2 V \longrightarrow \Lambda^2 V$$

satisfies $\star^2 = -I$. Every element of $SO^+(1, 3)$ preserves both the Lorentz metric and the chosen orientation, so $\star$ commutes with the proper Lorentz action.

After complexification,

$$W_4^{\mathbb{C}} = W_+ \oplus W_-, \quad W_{\pm} := \{B \in W_4^{\mathbb{C}} ; \star B = \pm iB\}. \quad (63)$$

The two spaces W_+ and W_- are three-dimensional irreducible complex Lorentz modules. They are inequivalent as complex $\mathfrak{so}(1, 3)$ -modules and correspond to the two simple factors in

$$\mathfrak{so}(4, \mathbb{C}) \simeq \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C}). \quad (64)$$

Let

$$T \in \text{End}_{G_4}(W_4).$$

Its complexification $T_{\mathbb{C}}$ is an intertwiner of $W_4^{\mathbb{C}}$. Because the decomposition $W_4^{\mathbb{C}} = W_+ \oplus W_-$ is multiplicity-free and the two summands are inequivalent irreducible modules, $T_{\mathbb{C}}$ preserves each summand. Schur's lemma therefore gives

$$T_{\mathbb{C}}|_{W_+} = \lambda_+ I, \quad T_{\mathbb{C}}|_{W_-} = \lambda_- I. \quad (65)$$

Complex conjugation exchanges W_+ and W_- . Since T is real,

$$\lambda_- = \overline{\lambda_+}. \quad (66)$$

Writing

$$\lambda_+ = a + ib, \quad a, b \in \mathbb{R},$$

gives $\lambda_- = a - ib$. Since $\star$ has eigenvalues $+i$ and $-i$ on W_+ and W_- , respectively, the corresponding real endomorphism is

$$T = aI + b\star. \quad (67)$$

Therefore

$$\text{End}_{G_4}(W_4) = \mathbb{R}I \oplus \mathbb{R}\star,$$

as claimed. $\square$

The four-dimensional commutant is therefore a real algebra canonically isomorphic, once an orientation has been chosen, to the complex numbers:

$$C_4 \simeq \mathbb{C}, \quad I \longleftrightarrow 1, \quad \star \longleftrightarrow i. \quad (68)$$

At this stage, Eq. (68) is an algebraic statement about the commutant of the Lorentz bivector representation. It is not yet the stronger physical statement that the complex structure of the quantum comparison sector must be represented by $\star$. That additional identification requires a compatibility condition between the quantum comparison carrier and the Lorentz spin cover and will be addressed in Section 3.

The commutant classification determines all possible Lorentz-equivariant real complex structures.

Theorem 2.3 (Adjoint-complexity rigidity). *Let $d \geq 2$. A Lorentz-equivariant real endomorphism*

$$J_G : \Lambda^2 \mathbb{R}^{1,d-1} \longrightarrow \Lambda^2 \mathbb{R}^{1,d-1}, \quad J_G^2 = -I \quad (69)$$

exists if and only if

$$d = 4.$$

In four dimensions there are exactly two such structures:

$$J_G = +\star \quad \text{or} \quad J_G = -\star. \quad (70)$$

Proof. For $d = 2$, $d = 3$, and $d \geq 5$, Theorem 2.2 gives

$$J_G = aI$$

for some $a \in \mathbb{R}$. Hence

$$J_G^2 = a^2 I,$$

which cannot equal $-I$.

For $d = 4$, write

$$J_G = aI + b\star, \quad a, b \in \mathbb{R}.$$

Using $\star^2 = -I$, one finds

$$J_G^2 = (a^2 - b^2)I + 2ab\star. \quad (71)$$

The condition $J_G^2 = -I$ is therefore equivalent to

$$2ab = 0 \quad (72)$$

and

$$a^2 - b^2 = -1. \quad (73)$$

If $b = 0$, then $a^2 = -1$, which has no real solution. Hence $a = 0$, and the remaining equation gives $b^2 = 1$. Therefore

$$J_G = \pm \star .$$

□

Theorem 2.3 gives a precise meaning to the exceptional character of four spacetime dimensions. The result is not merely the observation that the Hodge operator happens to satisfy $\star^2 = -I$ on Lorentzian bivectors in four dimensions. Rather, the full observer-equivariant endomorphism algebra of the bivector carrier has first been classified throughout the one-time Lorentz family. Within the class $d \geq 2$ considered here, only $d = 4$ admits an intrinsic Lorentz-equivariant real complex structure on W_d .

This motivates the following MS compatibility condition.

Assumption 2.4 (Gravitational realization of comparison complexity). *The coherent quantum-comparison structure possesses a real representation on the local gravitational generator sector*

$$W_d = \Lambda^2 V$$

through an observer-equivariant real-linear operator J_G satisfying

$$J_G^2 = -I.$$

This assumption should not be confused with ordinary complexification of the gravitational variables. It requires that the complex quarter-turn associated with coherent quantum comparison be represented intrinsically inside the real gravitational generator sector.

The dimensional consequence is immediate.

Corollary 2.5 (Conditional four-dimensional rigidity). *Within the generalized one-time Lorentz MS family, Assumption 2.4 implies*

$$d = 4 \quad (74)$$

and

$$J_G = \pm \star . \quad (75)$$

This is a conditional dimensional-selection theorem, not an assumption-free derivation of space-time dimension. Its logical content is

comparison complexity represented intrinsically on gravitational bivectors $\implies$ 3+1 dimensions.

If the quantum-comparison complex structure is not required to admit such a realization on the gravitational generator sector, the argument places no restriction on d .

The result is stronger than the elementary middle-degree observation that the Hodge operator maps $\Lambda^2 V$ to itself as another two-form only when the spacetime dimension is four. That observation

begins with the Hodge construction and asks when its source and target degrees coincide. The present argument instead begins with the representation-theoretic conditions

$$J_G \in \text{End}_{G_d}(W_d), \quad J_G^2 = -I,$$

classifies the complete equivariant endomorphism algebra, and only then identifies the surviving four-dimensional complex structure with Hodge duality.

The exceptional property can also be expressed directly at the Lie-algebra level. For $d = 4$,

$$\mathfrak{so}(1, 3) \simeq \mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}, \quad (76)$$

where the right-hand side denotes the underlying real Lie algebra of the complex simple Lie algebra $\mathfrak{sl}(2, \mathbb{C})$. Multiplication by the scalar complex unit on $\mathfrak{sl}(2, \mathbb{C})$, viewed as a real-linear map, commutes with its adjoint action. Through the equivariant identification $W_4 \simeq \mathfrak{so}(1, 3)$, this gives a Lorentz-equivariant real complex structure on bivectors, and Theorem 2.3 forces that structure to be $\pm\star$. Section 3 will make this relation explicit through the differential of the Lorentz spin covering map.

There is also a useful description in terms of the complexified bivector representation. In four dimensions,

$$W_4^{\mathbb{C}} = W_+ \oplus W_-, \quad (77)$$

with

$$\star B_{\pm} = \pm i B_{\pm}. \quad (78)$$

The corresponding spectral projectors are

$$P_{\pm} = \frac{1}{2} (I \mp i\star). \quad (79)$$

They satisfy

$$P_{\pm}^2 = P_{\pm}, \quad P_+ P_- = 0, \quad P_+ + P_- = I.$$

Because $\star$ is real, complex conjugation reverses its imaginary eigenvalue and therefore exchanges the two sectors:

$$\overline{W_+} = W_-. \quad (80)$$

Thus the real six-dimensional Lorentz bivector representation can equivalently be regarded as a three-complex-dimensional representation, with the opposite Hodge eigenspace providing its complex-conjugate description. The Hodge operator supplies the intrinsic complex multiplication.

This is the structure that fails in the other one-time Lorentz dimensions covered by Theorem 2.2. Their bivector representations admit complexification, and Lorentz groups in those dimensions may of course possess complex spinor or tensor representations. What fails is the more specific condition that the real adjoint/bivector carrier itself possess an observer-invariant endomorphism squaring to $-I$. Schematically,

$$\text{existence of complex representations} \not\Rightarrow \exists J_G \in \text{End}_{G_d}(W_d) \text{ with } J_G^2 = -I. \quad (81)$$

Nor does Theorem 2.3 determine the sign in $J_G = \pm\star$. Reversing the spacetime orientation changes $\star \mapsto -\star$, while complex conjugation interchanges the two Hodge eigenspaces. Selecting one sign therefore requires an orientation convention or an additional physical choice.

The theorem also does not determine a gravitational coupling constant. The operator J_G fixes an invariant complex direction in the gravitational response algebra but carries no dimensionful stiffness. If a constitutive operator introduced later takes the form

$$C = KJ_G,$$

then the magnitude K remains independent dynamical data. The distinction may be summarized as

$$\text{complex direction} \neq \text{response magnitude}. \quad (82)$$

Representation theory can constrain the former; it does not determine the latter.

A possible objection is that the argument uses the adjoint/bivector carrier rather than a larger gravitational phase space. This restriction is deliberate. In the first-order MS comparison structure, infinitesimal Lorentz transport is naturally carried by

$$\mathfrak{so}(1, d-1) \simeq \Lambda^2 V.$$

The bivector sector is therefore the minimal observer-covariant carrier shared by Lorentz generators and gravitational area-type response. A complex structure that becomes available only after adjoining arbitrary auxiliary degrees of freedom would not yield the same dimensional rigidity.

The bulk result may therefore be summarized as

$$\begin{aligned} \text{generalized MS dimension } d &\longrightarrow W_d = \Lambda^2 V, \\ \text{observer covariance} &\longrightarrow C_d = \text{End}_{G_d}(W_d), \\ J_G^2 = -I &\longrightarrow d = 4, \\ d = 4 &\longrightarrow J_G = \pm \star. \end{aligned} \quad (83)$$

This is the bulk complex-comparison rigidity result.

Two substantial questions remain. First, the mathematical uniqueness of the four-dimensional bivector complex structure does not by itself show that the complex structure appearing in the projective quantum sector must be represented by the same J_G . Section 3 addresses the representation-theoretic part of this question: if the quantum comparison carrier and the operational Lorentz geometry are required to use the same Lorentz spin cover, multiplication by i on the spin-cover Lie algebra is transported to $\pm\star$ on bivectors.

Second, it is useful to ask whether the same spacetime dimension can be selected by an independent argument. The projective null boundary provides such a criterion. In generalized spacetime dimension d , the sphere of null directions is S^{d-2} . Requiring a nontrivial rank-one helicity line bundle with nonzero first Chern class will independently force $d = 4$. This boundary argument is developed in the next section.

3 Spinorial Realization, Hodge Duality, and Null-Boundary Topology

Section 2 established a purely real representation-theoretic result. If the gravitational bivector sector is required to carry a Lorentz-equivariant real complex structure, then within the generalized

one-time Lorentz family the spacetime dimension is forced to be four, and the corresponding complex structure is

$$J_G = \pm \star .$$

That result did not require spinors.

We now address a complementary question. In four dimensions the proper orthochronous Lorentz group admits the spin covering

$$SL(2, \mathbb{C}) \longrightarrow SO^+(1, 3).$$

The Lie algebra $\mathfrak{sl}(2, \mathbb{C})$, regarded as a real six-dimensional Lie algebra, carries a canonical real-linear complex structure given by multiplication by i . We ask whether this complex structure is transported, through the spin-cover differential and the metric identification of Lorentz generators with bivectors, to the Hodge complex structure found in Section 2.

The distinction between the relevant operators is important. The scalar operator

$$\psi \longmapsto i\psi$$

on a complex spinor or quantum state space is not literally the same operator as

$$B \longmapsto \star B$$

on the real Lorentz bivector space. Nor is either of these operators literally identical to the real-linear map

$$X \longmapsto iX$$

on the underlying real Lie algebra $\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$. They act on different spaces.

The precise statement developed below is representation-theoretic: multiplication by i on $\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$ defines an $SL(2, \mathbb{C})$ -equivariant real complex structure, and its image under the Lorentz covering differential is exactly the Hodge complex structure on Lorentz bivectors, up to the orientation and sign conventions used in the identifications.

3.1 The spin-cover complex structure

Let

$$S_L \simeq \mathbb{C}^2 \tag{84}$$

denote the defining left-handed Weyl module of $SL(2, \mathbb{C})$. Its traceless complex endomorphisms form

$$\text{End}_0(S_L) \simeq \mathfrak{sl}(2, \mathbb{C}) \tag{85}$$

as a complex Lie algebra.

When regarded as a real Lie algebra,

$$\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$$

has real dimension six. The spin covering homomorphism

$$\pi : SL(2, \mathbb{C}) \longrightarrow SO^+(1, 3) \tag{86}$$

has kernel

$$\ker \pi = \{\pm I\},$$

and its differential at the identity is a real Lie-algebra isomorphism

$$d\pi : \mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}} \xrightarrow{\simeq} \mathfrak{so}(1, 3). \quad (87)$$

Using the Lorentz metric, we also have the equivariant real isomorphism

$$\mathfrak{so}(1, 3) \simeq \Lambda^2 V. \quad (88)$$

Whenever no ambiguity results, we absorb this metric identification into the notation $d\pi$, so that

$$d\pi : \mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}} \longrightarrow \Lambda^2 V$$

is understood as a real equivariant isomorphism.

The complex vector space $\mathfrak{sl}(2, \mathbb{C})$ possesses a canonical real-linear endomorphism

$$J_{\text{spin}} X := iX. \quad (89)$$

Since multiplication by i is complex linear,

$$J_{\text{spin}}^2 = -I. \quad (90)$$

Moreover, the adjoint representation of $SL(2, \mathbb{C})$ is complex linear, so

$$J_{\text{spin}} \text{Ad}_g = \text{Ad}_g J_{\text{spin}} \quad \text{for all } g \in SL(2, \mathbb{C}). \quad (91)$$

Transport J_{spin} to the Lorentz bivector representation by defining

$$J_B := d\pi \circ J_{\text{spin}} \circ d\pi^{-1}. \quad (92)$$

Then

$$J_B^2 = -I, \quad (93)$$

and J_B commutes with the proper Lorentz action.

The classification obtained in Section 2 therefore determines J_B completely.

Theorem 3.1 (Spin-cover–Hodge compatibility). *Let*

$$d\pi : \mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}} \xrightarrow{\simeq} \mathfrak{so}(1, 3) \simeq \Lambda^2 V$$

be the real differential of the Lorentz spin covering map, together with the metric identification of Lorentz generators and bivectors. Let

$$J_{\text{spin}} X = iX.$$

Then the induced real complex structure

$$J_B = d\pi J_{\text{spin}} d\pi^{-1}$$

on the Lorentz bivector representation satisfies

$$J_B = \pm \star. \quad (94)$$

For a fixed orientation and a fixed convention for the identification $d\pi$, the sign is fixed. Reversing the spacetime orientation reverses the sign of $\star$.

Proof. By construction,

$$J_B^2 = -I.$$

The equivariance of $d\pi$, together with Eq. (91), implies

$$J_B \in \text{End}_{SO^+(1,3)}(\Lambda^2 V).$$

By Theorem 2.2,

$$J_B = aI + b\star$$

for some $a, b \in \mathbb{R}$. The condition $J_B^2 = -I$ was solved in Theorem 2.3 and gives

$$a = 0, \quad b = \pm 1.$$

Hence

$$J_B = \pm \star.$$

□

The theorem gives the precise representation-theoretic content of the symbolic correspondence

$$i \longleftrightarrow \star. \quad (95)$$

The statement is not an equality of operators on one common vector space. Rather,

$$X \mapsto iX$$

on the real Lorentz spin-cover algebra is transported by the covering differential to

$$B \mapsto \pm \star B$$

on Lorentz bivectors.

This can be displayed explicitly in a standard basis. Let R_i and B_i denote rotation and boost generators of $\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$, chosen as

$$R_i = -\frac{i}{2}\sigma_i, \quad B_i = \frac{1}{2}\sigma_i, \quad (96)$$

where σ_i are the Pauli matrices. Then

$$J_{\text{spin}} R_i = B_i, \quad J_{\text{spin}} B_i = -R_i. \quad (97)$$

Choose the spacetime orientation and the corresponding bivector convention so that rotation-type and boost-type bivectors J_i, K_i satisfy

$$\star J_i = K_i, \quad \star K_i = -J_i. \quad (98)$$

If the covering differential is normalized by

$$d\pi(R_i) = J_i, \quad d\pi(B_i) = K_i, \quad (99)$$

then Eq. (97) becomes

$$d\pi J_{\text{spin}} d\pi^{-1} = \star.$$

With the opposite orientation the Hodge operator changes sign.

3.2 Chiral decomposition and the Clifford realization

The same structure appears in the complexified bivector representation. Let

$$V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}.$$

Then

$$\Lambda^2 V_{\mathbb{C}} = \Lambda_+^2 \oplus \Lambda_-^2, \quad (100)$$

where

$$\Lambda_{\pm}^2 := \{X \in \Lambda^2 V_{\mathbb{C}}; \star X = \pm iX\}. \quad (101)$$

Each summand is a three-dimensional irreducible complex Lorentz module, and complex conjugation exchanges them:

$$\overline{\Lambda_+^2} = \Lambda_-^2. \quad (102)$$

Let

$$S_R$$

denote the right-handed Weyl module, the complex-conjugate chirality to S_L . With a consistent assignment of chirality conventions,

$$\Lambda_+^2 \simeq \text{Sym}^2 S_L, \quad \Lambda_-^2 \simeq \text{Sym}^2 S_R, \quad (103)$$

or with $+$ and $-$ interchanged if the opposite Hodge/chirality convention is adopted.

Thus the two Hodge eigenspaces are precisely the two chiral complex Lorentz-generator representations.

This also explains why the statement

$$i = \star$$

would be mathematically imprecise. Scalar multiplication by i on a Weyl spinor space, multiplication by i on $\mathfrak{sl}(2, \mathbb{C})$, and the Hodge operator on bivectors are distinct operators.

Their relation may also be expressed through the Clifford representation. Let

$$\rho_{\text{spin}} : \mathfrak{so}(1, 3) \longrightarrow \text{End}(S_D)$$

denote the Dirac-spinor representation of the Lorentz algebra, where $S_D = S_L \oplus S_R$. For a fixed choice of metric, orientation, and Clifford conventions, there exists a convention sign $\varepsilon_{\star} = \pm 1$ such that

$$\rho_{\text{spin}}(\star X) = \varepsilon_{\star} i \gamma_5 \rho_{\text{spin}}(X). \quad (104)$$

On a Weyl subspace,

$$\gamma_5|_{S_L} = \varepsilon_L I, \quad \gamma_5|_{S_R} = -\varepsilon_L I,$$

with $\varepsilon_L = \pm 1$ depending on convention. Hence Eq. (104) reduces on each chirality to multiplication by one of the phases

$$+i \quad \text{or} \quad -i.$$

The important invariant statement is therefore not a literal identity of operators but the compatibility of the chiral complex structures carried by the same Lorentz representation theory.

This motivates the following physical assumption.

Assumption 3.2 (Same-spin-cover compatibility). *The projective quantum-comparison sector and the operational Lorentz geometry are realized through the same local Lorentz spin covering group*

$$\text{Spin}^+(1, 3) \simeq SL(2, \mathbb{C}), \quad (105)$$

and the infinitesimal Lorentz comparison generators entering the two descriptions are related through the same covering differential $d\pi$.

The assumption concerns the identification of the Lorentz symmetry representation. It does not assert that the spinorial Hilbert space and the bivector space are the same vector space, nor that scalar multiplication by i on the quantum carrier is literally the Hodge star.

Corollary 3.3 (Canonical transfer of the spin-cover complex structure). *Under Assumption 3.2, the canonical real complex structure*

$$X \mapsto iX$$

on $\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$ is represented on the gravitational bivector sector by

$$J_B = \pm \star. \quad (106)$$

Thus the spin-cover Lie algebra and the gravitational bivector representation carry two representation images of the same invariant real complex structure.

The remaining physical problem is dynamical. A nontrivial projective rotational comparison sector does not by itself prove that its relativistic completion must use the same $\text{Spin}^+(1, 3)$ cover as the operational soldering geometry. Localization, Lorentz covariance, and a relativistic continuum completion are natural ingredients in such an identification, but that derivation is not supplied here.

Accordingly, the logical implication established in this section is conditional:

common Lorentz spin-cover realization

$$\begin{aligned} + J_{\text{spin}} X &= iX \\ \implies J_B &= \pm \star. \end{aligned} \quad (107)$$

3.3 The Weyl carrier and the celestial sphere

The same two-complex-dimensional Weyl carrier also organizes several standard structures appearing elsewhere in the MS reconstruction.

Every Hermitian 2×2 matrix can be written uniquely as

$$X = x^0 I + x^i \sigma_i. \quad (108)$$

Its determinant is

$$\det X = (x^0)^2 - |\mathbf{x}|^2. \quad (109)$$

Thus the real vector space

$$\text{Herm}_2(\mathbb{C})$$

is naturally identified with Minkowski vector space $\mathbb{R}^{1,3}$, with determinant giving its Lorentz quadratic form.

The group $SL(2, \mathbb{C})$ acts by

$$X \mapsto gXg^\dagger. \quad (110)$$

This preserves $\det X$ and induces the usual proper orthochronous Lorentz action. The kernel of this action is $\{\pm I\}$, giving the covering map in Eq. (86).

The projective ray space of the Weyl module is

$$\mathbb{P}(S_L) \simeq \mathbb{CP}^1 \simeq S^2. \quad (111)$$

To see its relation with null directions, take a nonzero Weyl spinor

$$\lambda \in S_L \setminus \{0\}$$

and define

$$k(\lambda) := \lambda \lambda^\dagger. \quad (112)$$

This is a nonzero positive-semidefinite Hermitian matrix of rank one. Therefore

$$\det k(\lambda) = 0. \quad (113)$$

Under the identification $\text{Herm}_2(\mathbb{C}) \simeq \mathbb{R}^{1,3}$, it represents a future-pointing null vector.

If

$$\lambda \mapsto z\lambda, \quad z \in \mathbb{C}^\times,$$

then

$$k(z\lambda) = |z|^2 k(\lambda). \quad (114)$$

Thus the corresponding null *direction* depends only on the projective ray

$$[\lambda] \in \mathbb{CP}^1.$$

Conversely, every nonzero future null Hermitian matrix has rank one and can be written in the form

$$k = \lambda \lambda^\dagger$$

for a nonzero spinor λ , unique up to multiplication by a nonzero complex scalar after the overall null-vector normalization is discarded.

Hence there is a natural identification

$$\mathbb{CP}^1 \simeq \{\text{future null directions in } \mathbb{R}^{1,3}\} \simeq S^2. \quad (115)$$

This is an identification of homogeneous/projective spaces. It should not be interpreted as saying that a spinor ray and a null vector are literally the same mathematical object.

3.4 Generalized MS null boundary

The projective description connects naturally with the generalized MS compact-state boundary. Let

$$D \in \mathbb{R}^m, \quad |D|^2 + S^2 = 1, \quad S \geq 0. \quad (116)$$

The interior

$$S > 0$$

is parametrized by

$$|D| < 1.$$

The null or projective boundary occurs at

$$S = 0, \quad |D| = 1.$$

Therefore the space of boundary directions is

$$\partial B^m \simeq S^{m-1}. \quad (117)$$

Since

$$d = m + 1,$$

the generalized null-direction sphere is

$$\mathcal{N}_d \simeq S^{d-2}. \quad (118)$$

For $d = 4$, this is precisely

$$S^2 \simeq \mathbb{CP}^1.$$

3.5 Helicity line bundles and first Chern class

In four-dimensional relativistic theory, a fixed nonzero null direction has little group

$$ISO(2).$$

For the ordinary finite-helicity representations in which the translational part of the little group acts trivially, the physical one-dimensional polarization character is a $U(1)$ representation. Varying the null direction over the celestial sphere produces a complex line bundle

$$L_h \longrightarrow S^2 \quad (119)$$

for a fixed helicity h .

For the standard helicity line construction, with $2h \in \mathbb{Z}$, the first Chern number is

$$\langle c_1(L_h), [S^2] \rangle = -2h, \quad (120)$$

up to the simultaneous choices of sphere orientation and helicity convention. Reversing these conventions changes the sign but not the absolute value.

The result Eq. (120) is a four-dimensional helicity statement. It should not be assumed without modification for arbitrary higher-dimensional massless little-group representations.

We now elevate one specific feature of this construction to a conditional dimensional requirement.

Assumption 3.4 (Rank-one helicity topology). *In the generalized d -dimensional MS null sector, a fundamental finite-helicity degree of freedom is represented over the null-direction sphere by a complex line bundle*

$$L \longrightarrow S^{d-2} \quad (121)$$

whose first Chern class is nonzero:

$$c_1(L) \neq 0. \quad (122)$$

This is an additional physical assumption, not a consequence of the existence of massless states alone. It specifically requires the generalized boundary sector to retain an ordinary rank-one complex $U(1)$ -type helicity topology.

The topology of spheres then yields an independent dimensional constraint.

Theorem 3.5 (Helicity–Chern dimensional rigidity). *Under Assumption 3.4, the spacetime dimension is*

$$d = 4. \quad (123)$$

Proof. The first Chern class of a complex line bundle

$$L \longrightarrow S^{d-2}$$

is an element of

$$c_1(L) \in H^2(S^{d-2}; \mathbb{Z}). \quad (124)$$

The integral cohomology of a sphere satisfies

$$H^2(S^n; \mathbb{Z}) \simeq \begin{cases} \mathbb{Z}, & n = 2, \\ 0, & n \neq 2. \end{cases} \quad (125)$$

Therefore

$$c_1(L) \neq 0$$

is possible only when

$$d - 2 = 2.$$

Hence

$$d = 4.$$

□

The theorem is mathematically independent of Theorem 2.3. The bulk theorem is a statement about the real commutant of the Lorentz adjoint representation, whereas Theorem 3.5 is a statement about the second integral cohomology of the null-direction sphere.

The two arguments therefore use different mathematical inputs.

Corollary 3.6 (Bulk–boundary four-dimensional double rigidity). *Consider the generalized one-time Lorentz MS family.*

If the following two independent requirements are imposed:

1. *coherent comparison complexity is represented by a Lorentz-equivariant real complex structure on the gravitational bivector carrier; and*
2. *the fundamental null-boundary finite-helicity sector is represented by a complex line bundle over S^{d-2} with nonzero first Chern class,*

then each requirement independently implies

$$d = 4. \quad (126)$$

The logical structure is therefore

$$\begin{aligned} \text{bulk adjoint-complex condition} &\implies d = 4, \\ \text{boundary rank-one Chern condition} &\implies d = 4. \end{aligned} \quad (127)$$

Their agreement provides a bulk–boundary consistency between two mathematically independent conditional mechanisms.

The assumptions must remain visible. In dimensions greater than four, massless little groups possess polarization representations that need not reduce to one-dimensional $U(1)$ helicity characters. Higher-rank complex bundles, non-Abelian Berry connections, and other polarization structures are not excluded by Theorem 3.5. The theorem says only that the particular mechanism

nontrivial complex helicity line over the full null-direction sphere

with

$$c_1 \neq 0$$

requires that sphere to be S^2 .

3.6 The minimal helicity line and the Hopf bundle

In four dimensions there is a further standard correspondence. Suppose that the minimal projective carrier under consideration has

$$|h| = \frac{1}{2}. \quad (128)$$

Then Eq. (120) gives

$$\left| \langle c_1(L_{1/2}), [S^2] \rangle \right| = 1. \quad (129)$$

For one choice of orientation and helicity convention,

$$c_1(L_{1/2}) = -1.$$

Since

$$S^2 \simeq \mathbb{CP}^1,$$

isomorphism classes of complex line bundles over S^2 are classified by their first Chern class. Under the convention

$$c_1(L_{1/2}) = -1,$$

one therefore has

$$L_{1/2} \simeq \mathcal{O}(-1), \quad (130)$$

where

$$\mathcal{O}(-1) \longrightarrow \mathbb{CP}^1$$

is the tautological complex line bundle. With the opposite orientation or helicity convention one obtains the dual bundle $\mathcal{O}(1)$.

The tautological bundle has fiber

$$\mathcal{O}(-1)_{[\lambda]} = \mathbb{C}\lambda \subset S_L \quad (131)$$

over the projective point $[\lambda] \in \mathbb{P}(S_L)$. Its unit-circle bundle is the Hopf fibration

$$S^3 \longrightarrow S^2.$$

Thus, when the minimal helicity sector has $|h| = 1/2$, the helicity line bundle over the celestial sphere and the tautological ray bundle of a two-component complex spinor have the same line-bundle isomorphism class, up to dualization associated with orientation and helicity conventions.

This is stronger than the statement that their base spaces are both homeomorphic to S^2 , but it is still an isomorphism of bundle types, not an ontological identification of the physical constructions.

3.7 Conditional spinorial–boundary closure

The standard mathematical correspondences appearing in this section may now be summarized as

$$S_L \simeq \mathbb{C}^2 \implies \mathbb{P}(S_L) \simeq \mathbb{CP}^1 \simeq S^2, \quad (132)$$

$$\text{Herm}_2(\mathbb{C}) \simeq \mathbb{R}^{1,3}, \quad \det X = \eta(X, X), \quad (133)$$

$$\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}} \xrightarrow{d\pi} \mathfrak{so}(1, 3) \simeq \Lambda^2 V, \quad (134)$$

$$X \mapsto iX \xrightarrow{d\pi} J_B = \pm \star, \quad (135)$$

and, for the standard minimal half-helicity construction,

$$L_{1/2} \simeq \mathcal{O}(\mp 1) \longrightarrow \mathbb{CP}^1, \quad (136)$$

where the sign depends on orientation and helicity convention.

Most of these mathematical correspondences are standard. The physical question specific to the MS reconstruction is whether a single dynamically selected comparison structure is responsible for these different representation-theoretic appearances, rather than the same two-dimensional complex carrier being introduced independently in different parts of the construction.

The logical content of the present section should therefore be separated into three levels.

First, there is an exact representation-theoretic statement:

$$J_{\text{spin}} X = iX \implies d\pi J_{\text{spin}} d\pi^{-1} = \pm \star. \quad (137)$$

Second, there is an independent topological implication:

$$c_1(L) \neq 0 \text{ for a complex line bundle } L \rightarrow S^{d-2} \implies d = 4. \quad (138)$$

Third, there is a physical compatibility assumption: the projective quantum-comparison carrier and the reconstructed Lorentz geometry are required to realize the same Lorentz spin cover. The present paper does not derive this common-spin-cover identification dynamically.

Within these qualifications, the results of this section establish the spinorial and null-boundary side of the proposed complex-comparison closure. In the bulk, the invariant complex structure of the Lorentz spin-cover algebra is represented on gravitational bivectors by Hodge duality. At the null boundary, the standard rank-one helicity topology selects the same spacetime dimension and, in the minimal half-helicity case, reproduces the tautological Hopf line-bundle class over $\mathbb{CP}^1 \simeq S^2$.

A complex structure on the gravitational generator carrier does not, however, determine a gravitational constitutive law or its magnitude. The next section turns to this separate problem. Four-dimensional Lorentz covariance restricts a local real linear constitutive operator on bivectors to

$$\mathbb{R}I \oplus \mathbb{R}\star,$$

and a further comparison-complex compatibility condition will be used to select the pure Hodge constitutive direction.

4 Complex Constitutive Response and the Einstein Direction

The preceding sections determine a distinguished real complex structure on the four-dimensional Lorentz bivector carrier. They do not, however, determine a gravitational constitutive law. A complex structure specifies an invariant direction in the algebra of Lorentz-equivariant response operators, whereas a constitutive law must additionally specify the magnitude and orientation with which a geometric field-strength variable is converted into its corresponding gravitational excitation.

We now make this distinction explicit.

Let V be an oriented and time-oriented four-dimensional Lorentz vector space and define

$$W := \Lambda^2 V. \quad (139)$$

For the chosen orientation and Lorentz metric, the Hodge operator

$$\star : W \longrightarrow W$$

satisfies

$$\star^2 = -I. \quad (140)$$

In first-order gravitational language, the solder field determines an area-type bivector-valued two-form

$$B = e \wedge e. \quad (141)$$

The spacetime differential-form indices will be suppressed below; the constitutive operator considered in this section acts pointwise on the internal bivector carrier W .

Let H denote the corresponding gravitational excitation. A local, real-linear constitutive relation has the form

$$H = C(B), \quad (142)$$

where

$$C : W \longrightarrow W. \quad (143)$$

If the local constitutive response is required to commute with the proper orthochronous Lorentz action, then

$$C \in \text{End}_{SO^+(1,3)}(W). \quad (144)$$

By Theorem 2.2,

$$C = \alpha I + \beta \star, \quad \alpha, \beta \in \mathbb{R}. \quad (145)$$

Thus the most general local real-linear Lorentz-equivariant constitutive operator on the internal bivector carrier is described by two real coefficients. The relevant algebra is

$$\text{End}_{SO^+(1,3)}(W) = \mathbb{R}I \oplus \mathbb{R}\star \simeq \mathbb{C}, \quad I \leftrightarrow 1, \quad \star \leftrightarrow i. \quad (146)$$

For a nonzero constitutive operator,

$$(\alpha, \beta) \neq (0, 0),$$

define

$$K := \sqrt{\alpha^2 + \beta^2} > 0. \quad (147)$$

Choose an angle

$$\theta \in \mathbb{R}/2\pi\mathbb{Z}$$

such that

$$\alpha = K \cos \theta, \quad \beta = K \sin \theta. \quad (148)$$

Equivalently,

$$\theta = \text{atan2}(\beta, \alpha) \pmod{2\pi}.$$

Since $\star^2 = -I$,

$$e^{\theta\star} = \cos \theta I + \sin \theta \star. \quad (149)$$

Hence

$$C = K (\cos \theta I + \sin \theta \star) = K e^{\theta\star}. \quad (150)$$

This decomposition separates two logically distinct constitutive data.

Definition 4.1 (Gravitational constitutive stiffness and phase). *For a nonzero local Lorentz-equivariant constitutive operator*

$$C = \alpha I + \beta \star,$$

the positive quantity

$$K = \sqrt{\alpha^2 + \beta^2} \quad (151)$$

will be called its constitutive stiffness, while

$$\theta = \text{atan2}(\beta, \alpha) \pmod{2\pi} \quad (152)$$

will be called its constitutive phase.

The word “phase” has a direct representation-theoretic meaning. After complexification,

$$W_{\mathbb{C}} = W_+ \oplus W_-, \quad \star X_{\pm} = \pm i X_{\pm}. \quad (153)$$

Therefore

$$C X_+ = (\alpha + i\beta) X_+ = K e^{i\theta} X_+, \quad (154)$$

$$C X_- = (\alpha - i\beta) X_- = K e^{-i\theta} X_-. \quad (155)$$

Thus a real Lorentz-equivariant constitutive law acts on the two chiral bivector sectors by complex-conjugate response factors.

The stiffness K and the phase θ have different logical status. Lorentz covariance permits both. The complex-structure rigidity established in the preceding sections identifies the distinguished endomorphism $\star$, but it does not by itself imply

$$\theta = \pm \frac{\pi}{2},$$

nor does it determine the magnitude K .

4.1 Einstein–Holst response and the pure Hodge direction

For comparison with the standard first-order gravitational parametrization, consider a constitutive operator of Einstein–Holst form

$$C_{\text{EH}} = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right), \quad (156)$$

where κ is the gravitational normalization and γ is the Holst or Immirzi parameter. Precise numerical factors depend on the normalization convention used for the action.

Comparing Eq. (156) with Eq. (145) gives

$$\alpha = \frac{1}{2\kappa\gamma}, \quad \beta = \frac{1}{2\kappa}. \quad (157)$$

Consequently,

$$K = \frac{1}{2|\kappa|} \sqrt{1 + \frac{1}{\gamma^2}}, \quad (158)$$

with the remaining overall sign absorbed into the phase convention, and, when the relevant denominators are nonzero,

$$\frac{\alpha}{\beta} = \frac{1}{\gamma} = \cot \theta. \quad (159)$$

The Einstein–Cartan constitutive direction is obtained by removing the identity channel:

$$\alpha = 0. \quad (160)$$

Then

$$C = \beta \star. \quad (161)$$

For the positive-stiffness convention,

$$K = |\beta|,$$

the two possible orientations of this ray correspond to

$$\theta = +\frac{\pi}{2} \quad \text{or} \quad -\frac{\pi}{2} \pmod{2\pi}. \quad (162)$$

Thus the Einstein–Cartan response occupies the pure Hodge ray in the two-dimensional Lorentz-equivariant constitutive plane:

$$\text{Einstein–Cartan direction} \quad \longleftrightarrow \quad \text{pure Hodge constitutive response}. \quad (163)$$

A finite identity component, as in a finite Holst contribution, displaces the constitutive map away from this pure Hodge ray and can therefore be described as a detuning of the constitutive phase.

This interpretation concerns the response operator itself. Whether the two channels are operationally distinguishable depends on the available sources and observables. In particular, spin–torsion response provides a natural channel in which the identity and Hodge components can contribute differently. The constitutive phase should therefore be regarded as a response parameter rather than merely as a rewriting of bare action coefficients.

4.2 Complex structure on the doubled constitutive response space

The polar form

$$C = Ke^{\theta\star}$$

raises a stronger question. Can the pure Hodge direction be selected by a compatibility condition associated with the complex structure of quantum comparison?

To formulate such a condition, introduce the doubled real constitutive response space

$$\mathcal{R}_G := W \oplus W. \quad (164)$$

Its elements will be written as

$$(B, H).$$

The two components need not carry the same physical units. Let

$$\zeta \in \mathbb{R} \setminus \{0\}$$

be a nonzero conversion parameter with the dimensions required to convert the units of H into those of B . Define

$$J_\zeta : \mathcal{R}_G \longrightarrow \mathcal{R}_G \quad (165)$$

by

$$J_\zeta(B, H) = (\zeta H, -\zeta^{-1}B). \quad (166)$$

A direct calculation gives

$$J_\zeta^2 = -I. \quad (167)$$

Thus J_ζ defines a real complex structure on the doubled constitutive response space.

The constitutive relation

$$H = CB$$

defines the graph

$$\Gamma_C := \{(B, H) \in W \oplus W ; H = CB\} = \{(B, CB) ; B \in W\}. \quad (168)$$

We may therefore ask whether the physical constitutive graph is invariant under the doubled quarter-turn J_ζ .

Theorem 4.2 (Complex-graph constitutive rigidity). *Let*

$$C : W \longrightarrow W$$

be a real-linear map and let

$$J_\zeta(B, H) = (\zeta H, -\zeta^{-1}B), \quad \zeta \neq 0.$$

Then

$$J_\zeta \Gamma_C = \Gamma_C \quad (169)$$

if and only if

$$C^2 = -\zeta^{-2}I. \quad (170)$$

Proof. Take any point

$$(B, CB) \in \Gamma_C.$$

Its image is

$$J_\zeta(B, CB) = (\zeta CB, -\zeta^{-1}B). \quad (171)$$

This point belongs to Γ_C precisely when its second component is obtained by applying C to its first:

$$-\zeta^{-1}B = C(\zeta CB) = \zeta C^2 B. \quad (172)$$

Since this must hold for every $B \in W$,

$$C^2 = -\zeta^{-2}I.$$

Conversely, suppose

$$C^2 = -\zeta^{-2}I.$$

Then for every B ,

$$C(\zeta CB) = \zeta C^2 B = -\zeta^{-1}B,$$

so

$$J_\zeta(B, CB) \in \Gamma_C.$$

Hence

$$J_\zeta \Gamma_C \subseteq \Gamma_C.$$

Because J_ζ is invertible and $J_\zeta^{-1} = -J_\zeta$, the reverse inclusion follows as well. Therefore

$$J_\zeta \Gamma_C = \Gamma_C.$$

□

Theorem 4.2 does not require Lorentz covariance. Once Lorentz covariance is imposed, however, the condition becomes highly restrictive.

Theorem 4.3 (Lorentz-covariant quarter-turn selection). *Let*

$$C = \alpha I + \beta \star, \quad \alpha, \beta \in \mathbb{R},$$

be a real Lorentz-equivariant constitutive operator on four-dimensional Lorentz bivectors. If

$$J_\zeta \Gamma_C = \Gamma_C,$$

then

$$C = \pm \zeta^{-1} \star. \quad (173)$$

Proof. Using $\star^2 = -I$,

$$C^2 = (\alpha^2 - \beta^2)I + 2\alpha\beta \star. \quad (174)$$

By Theorem 4.2, graph invariance requires

$$C^2 = -\zeta^{-2}I.$$

Since I and $\star$ are linearly independent over $\mathbb{R}$, we obtain

$$2\alpha\beta = 0 \quad (175)$$

and

$$\alpha^2 - \beta^2 = -\zeta^{-2}. \quad (176)$$

If $\beta = 0$, then

$$\alpha^2 = -\zeta^{-2},$$

which has no real solution. Therefore

$$\alpha = 0.$$

The remaining equation gives

$$\beta^2 = \zeta^{-2},$$

hence

$$\beta = \pm\zeta^{-1},$$

and therefore

$$C = \pm\zeta^{-1} \star.$$

□

Thus, within the local real-linear Lorentz-equivariant constitutive plane, the doubled graph-invariance condition eliminates the identity channel and selects the pure Hodge ray:

$$\begin{aligned} \text{Lorentz covariance} &\implies C = \alpha I + \beta \star, \\ J_\zeta \Gamma_C = \Gamma_C &\implies C^2 = -\zeta^{-2} I, \\ \text{both conditions} &\implies C = \pm\zeta^{-1} \star. \end{aligned} \quad (177)$$

The algebraic result is exact. Its physical use requires an additional compatibility assumption.

Assumption 4.4 (Comparison–constitutive compatibility). *The universal complex structure associated with coherent quantum comparison admits a faithful real realization on the doubled gravitational constitutive response space*

$$\mathcal{R}_G = W \oplus W$$

through an operator of the form

$$J_\zeta(B, H) = (\zeta H, -\zeta^{-1} B),$$

with $\zeta \neq 0$, and the physical constitutive relation is invariant under this quarter-turn:

$$J_\zeta \Gamma_C = \Gamma_C. \quad (178)$$

This assumption is not implied by Lorentz covariance and is not derived solely from the original MS operational reconstruction. It is an additional physical compatibility condition relating quantum comparison complexity to the gravitational constitutive response.

Under this assumption, the pure Hodge direction follows.

Corollary 4.5 (Conditional Einstein constitutive selection). *Under Lorentz covariance and Assumption 4.4, the constitutive operator lies on the pure Hodge ray:*

$$C = \varepsilon_C K_G \star, \quad \varepsilon_C = \pm 1, \quad (179)$$

where

$$K_G := |\zeta|^{-1} > 0. \quad (180)$$

The constitutive phase is therefore

$$\theta = \varepsilon_C \frac{\pi}{2} \pmod{2\pi}, \quad (181)$$

after the sign of ζ has been absorbed into the branch ε_C .

The distinction between ε_C and K_G is useful. The positive quantity K_G measures the magnitude of the constitutive response, while ε_C records the orientation or branch of the Hodge quarter-turn.

The result is the constitutive analogue of the complex-structure rigidity of the preceding sections. There, the assumption that comparison complexity is represented on gravitational bivectors led to the unique invariant complex directions

$$\pm \star .$$

Here, the additional requirement that the constitutive graph itself be closed under the doubled quarter-turn forces the constitutive map onto the same Hodge ray.

4.3 Logical status of the doubled quarter-turn

The meaning of J_ζ requires care. The doubled space

$$W \oplus W$$

introduced above is a constitutive response space. It should not be identified automatically with the canonical gravitational phase space.

In a first-order gravitational formulation, the gravitational excitation is dynamically paired with connection variables, while

$$B = e \wedge e$$

is constructed from the solder field. The pair

$$(B, H)$$

therefore represents the input and output sides of a constitutive law. It is not, merely by virtue of this construction, a canonical coordinate–momentum pair.

Accordingly,

$$J_\zeta^2 = -I$$

does not by itself establish a Kähler structure on the physical gravitational phase space, nor does it determine the gravitational symplectic form.

The content of Assumption 4.4 is narrower: it assumes that the complex quarter-turn associated with coherent comparison is represented on the constitutive variables by the exchange form

$$(B, H) \longmapsto (\zeta H, -\zeta^{-1} B).$$

Why this particular representation should follow from a deeper comparison theory remains a separate question. The form of J_ζ is compatible with three desired properties:

1. it exchanges the constitutive input and excitation variables;
2. it is invertible;
3. after inserting the dimensional conversion ζ , it satisfies $J_\zeta^2 = -I$.

These properties show that the construction is natural, but they do not by themselves prove uniqueness from more primitive MS axioms.

A microscopic derivation would have to explain why covariance, comparison complexity, and response duality imply this exchange structure rather than merely allowing it.

When

$$C = \varepsilon_C K_G \star,$$

the graph

$$\Gamma_C$$

is a complex subspace of $(\mathcal{R}_G, J_\zeta)$, provided the branch and conversion convention are chosen consistently. In this precise sense the constitutive law is closed under the same quarter-turn structure that exchanges its two response components.

4.4 Constitutive direction versus constitutive stiffness

The graph-rigidity argument selects a constitutive direction but does not supply a dynamical positivity theorem.

The sign or branch

$$\varepsilon_C = \pm 1$$

cannot be selected by Lorentz covariance and graph invariance alone. Likewise, positivity should not be imposed naively on the full covariant Lorentz bivector space, whose natural bilinear forms are indefinite.

Any physical positivity condition must instead be formulated on an appropriate reduced propagating or source-response sector, or after a suitable Hamiltonian, static, or Euclidean reduction.

Thus the present result establishes

$$C \propto \star \tag{182}$$

but does not by itself establish the stability condition

$$K_G > 0$$

from first principles. In the notation of Eq. (179), positivity of K_G is a convention for the magnitude; physical stability of the selected branch requires separate dynamical information.

Nor does the argument determine the numerical value of K_G . The conversion scale ζ remains dimensionful data whose value is not fixed by representation theory.

The logical separation is therefore

$$\text{complex comparison compatibility} \implies \text{constitutive direction}, \tag{183}$$

whereas

$$\text{gravitational response dynamics} \implies \text{constitutive stiffness}. \tag{184}$$

A generic local Lorentz-equivariant response

$$C = K e^{\theta \star}$$

therefore contains two distinct questions:

Which constitutive direction is selected?

and

What is the magnitude of the gravitational response?

The comparison-complex condition addresses the first. The second remains dynamical.

4.5 Scale dependence and the source-response map

The distinction becomes especially important in an effective description. At a scale μ , the leading local constitutive channel may take the form

$$C_{\text{eff}}(\mu) = \alpha(\mu)I + \beta(\mu)\star = K(\mu)e^{\theta(\mu)\star}, \quad (185)$$

provided the effective response has been projected onto the same local Lorentz-equivariant algebraic channel.

Even if the pure Hodge direction is approached in the infrared,

$$\theta(\mu) \longrightarrow \varepsilon_C \frac{\pi}{2}, \quad (186)$$

the stiffness

$$K(\mu)$$

may run independently. Neither the local commutant classification nor the quarter-turn argument proves that a microscopic renormalization flow remains within this two-dimensional constitutive truncation. Higher derivatives, nonlocal terms, additional torsional operators, or other response structures may appear. The effective closure problem will be treated separately in Section 7.

The local constitutive structure also determines the response to matter spin. Schematically, let

$$\tau$$

denote the matter spin source and let

$$\mathcal{T}_e$$

denote the solder-dependent real-linear map relating torsion T to the bivector-valued connection equation. The algebraic source equation may be written as

$$C \mathcal{T}_e(T) = \tau. \quad (187)$$

For

$$C = \alpha I + \beta \star$$

with

$$\alpha^2 + \beta^2 \neq 0,$$

the inverse constitutive operator is

$$C^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}. \quad (188)$$

Indeed,

$$(\alpha I + \beta \star)(\alpha I - \beta \star) = (\alpha^2 + \beta^2)I.$$

If $\mathcal{T}_e$ is invertible on the source-response channel under consideration, then

$$T = \mathcal{T}_e^{-1} C^{-1}(\tau). \quad (189)$$

The two constitutive channels therefore produce different source-to-torsion response tensors. This is why a spinful matter sector can, in principle, probe the constitutive orientation even in situations in which the spinless classical vacuum equations do not distinguish the two terms.

For the pure Hodge branch

$$C = \varepsilon_C K_G \star, \quad \varepsilon_C = \pm 1, \quad (190)$$

one has

$$C^{-1} = -\varepsilon_C K_G^{-1} \star, \quad (191)$$

because

$$\star^{-1} = -\star.$$

Hence

$$T = -\varepsilon_C K_G^{-1} \mathcal{T}_e^{-1} \star (\tau). \quad (192)$$

Thus the gravitational stiffness enters inversely in the algebraic spin–torsion susceptibility.

It is useful to define the corresponding classical response map

$$L_e := \mathcal{T}_e^{-1} C^{-1}, \quad (193)$$

so that

$$T = L_e \tau. \quad (194)$$

This prepares the quantum-response analysis of the next section. Once e , C , and the relevant inverse response channel have been fixed classically, the matter source can be promoted to an operator and its correlation structure can be transferred through L_e to an operator-valued torsional response.

The important logical separation is

$$\begin{aligned} \text{Lorentz representation theory} &\implies C \in \mathbb{R}I \oplus \mathbb{R}\star, \\ \text{comparison–constitutive compatibility} &\implies C \propto \star, \\ \text{response dynamics} &\implies K_G, \\ \text{quantum source structure} &\implies \text{quantum statistics of the induced response.} \end{aligned} \quad (195)$$

These steps should not be conflated. In particular, the complex quarter-turn does not by itself quantize the gravitational field, and the selection of the pure Hodge constitutive direction does not determine Newton’s coupling. The next section studies what can nevertheless be said when the classical spin source in Eq. (194) is replaced by a quantum operator.

5 Quantum Source Transduction and Correlated Geometric Response

The constitutive analysis of Section 4 determines how a classical spin source is converted into torsional geometry. We now ask what becomes of this relation when the matter source is quantum.

The distinction is important. A quantum matter source coupled through a fixed classical gravitational response kernel does not imply that the gravitational field has been quantized independently. Nevertheless, if a geometric response observable is defined as a deterministic linear image of a quantum source operator, then that geometric observable is itself operator valued and inherits quantum expectation values, correlations, commutators, and higher cumulants from the source.

This section develops that source-induced operator geometry and keeps it separate from intrinsic quantum gravitational degrees of freedom.

5.1 Classical spin–torsion response

Let e be a fixed nondegenerate solder field and let

$$C = \alpha I + \beta \star \quad (196)$$

be an invertible Lorentz-equivariant constitutive operator on the bivector response channel under consideration.

Let T denote torsion and let τ denote the matter spin source. As in the preceding section, write the relevant algebraic connection equation schematically as

$$C \mathcal{T}_e(T) = \tau, \quad (197)$$

where $\mathcal{T}_e$ is the solder-dependent real-linear map determined by the tensor structure of the connection equation.

The precise index conventions depend on the normalization of the first-order gravitational theory. The results below require only that C and $\mathcal{T}_e$ be invertible on the physical source-response channel being considered.

Define

$$L_e := \mathcal{T}_e^{-1} C^{-1}. \quad (198)$$

Then

$$T = L_e \tau. \quad (199)$$

For

$$C = \alpha I + \beta \star, \quad \alpha^2 + \beta^2 \neq 0,$$

one has

$$C^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}, \quad (200)$$

and hence

$$L_e = \mathcal{T}_e^{-1} \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}. \quad (201)$$

For the pure Hodge branch,

$$C = \varepsilon_C K_G \star, \quad \varepsilon_C = \pm 1, \quad K_G > 0, \quad (202)$$

we obtain

$$C^{-1} = -\varepsilon_C K_G^{-1} \star \quad (203)$$

and therefore

$$L_e = -\varepsilon_C K_G^{-1} \mathcal{T}_e^{-1} \star. \quad (204)$$

Thus the gravitational stiffness enters inversely in the algebraic spin–torsion susceptibility.

Within this invertible algebraic response sector,

$$\tau = 0 \implies T = 0,$$

and, because L_e is invertible on the selected channel,

$$T = 0 \implies \tau = 0.$$

These implications concern only the response channel on which the invertibility assumptions hold.

5.2 Source-induced torsion operators

Local quantum fields are operator-valued distributions. It is therefore useful to formulate the quantum response first in smeared form.

Let f be a smooth compactly supported test field of the tensor type dual to torsion and let

$$\widehat{\tau}(h) \quad (205)$$

denote the matter spin-source operator smeared against an appropriate test field h .

The classical linear response map L_e induces a transpose action on test fields. We denote the corresponding formal transpose by $L_e^\dagger$. Define the smeared torsion observable by

$$\widehat{T}(f) := \widehat{\tau}(L_e^\dagger f). \quad (206)$$

This formula is the distributionally precise meaning of the symbolic local relation

$$\widehat{T} = L_e \widehat{\tau}. \quad (207)$$

Equation (207) is not an independent canonical quantization of the torsion field. It defines the torsional geometric observable as the image of an already quantum source under a fixed c-number gravitational response map.

Definition 5.1 (Source-induced operator geometry). *A geometric observable $\widehat{G}$ will be called source induced or slaved on a given effective sector if it is completely determined by matter-source operators through a fixed state-independent c-number linear map,*

$$\widehat{G} = L\widehat{J}, \quad (208)$$

with no independent geometric operator algebra required to define $\widehat{G}$.

The torsion operator Eq. (206) is the primary example.

Proposition 5.2 (Expectation-value transduction). *Let*

$$\widehat{T} = L_e \widehat{\tau}$$

be source induced. Then, for every state for which the corresponding expectation values exist,

$$\langle \widehat{T} \rangle = L_e \langle \widehat{\tau} \rangle. \quad (209)$$

Thus the classical response equation survives directly at the level of one-point functions.

Define the centered operators

$$\Delta \widehat{\tau} = \widehat{\tau} - \langle \widehat{\tau} \rangle, \quad \Delta \widehat{T} = \widehat{T} - \langle \widehat{T} \rangle. \quad (210)$$

Then

$$\Delta \widehat{T} = L_e \Delta \widehat{\tau}. \quad (211)$$

For example, define the symmetrized source covariance by

$$C_{\tau\tau}(x, y) := \frac{1}{2} \langle \{ \Delta \widehat{\tau}(x), \Delta \widehat{\tau}(y) \} \rangle, \quad (212)$$

and similarly

$$C_{TT}(x, y) := \frac{1}{2} \langle \{ \Delta \widehat{T}(x), \Delta \widehat{T}(y) \} \rangle. \quad (213)$$

For a local state-independent response map,

$$C_{TT}(x, y) = L_x C_{\tau\tau}(x, y) L_y^T, \quad (214)$$

with the tensor contractions understood. For a nonlocal response kernel, the same statement holds with the corresponding integral contractions.

5.3 Connected correlator transduction

Higher correlation functions require an operator-ordering convention. Choose once and for all a multilinear ordering prescription O , such as a specified Wightman ordering, time ordering, or another ordering appropriate to the observable algebra.

Write

$$\langle O \widehat{O}_1 \cdots \widehat{O}_n \rangle_c \quad (215)$$

for the corresponding connected n -point functional, defined from the ordered moments by the usual moment–cumulant recursion.

The important property used below is multilinearity in each operator argument.

Theorem 5.3 (Quantum source-transduction theorem). *Suppose that a geometric operator is source induced through a deterministic state-independent c-number linear map,*

$$\widehat{T}(x) = L_x \widehat{\tau}(x),$$

or through the corresponding linear integral kernel in the nonlocal case.

For any fixed multilinear ordering prescription O ,

$$\begin{aligned} & \langle O \widehat{T}(x_1) \cdots \widehat{T}(x_n) \rangle_c \\ &= \left(\bigotimes_{r=1}^n L_{x_r} \right) \langle O \widehat{\tau}(x_1) \cdots \widehat{\tau}(x_n) \rangle_c, \end{aligned} \quad (216)$$

for every n for which the corresponding smeared connected correlators are well defined.

Proof. The connected correlator associated with a fixed linear ordering prescription is multilinear in its operator arguments. Substituting

$$\widehat{T}(x_r) = L_{x_r} \widehat{\tau}(x_r)$$

in every argument and using that the L_{x_r} are state-independent c-number linear maps allows each L_{x_r} to be pulled outside the correlator. This gives Eq. (216). $\square$

Thus the complete connected hierarchy associated with any specified ordering prescription is transferred through the classical linear response map.

The result should not be interpreted as saying that different operator orderings are equivalent. Time-ordered, Wightman, retarded, symmetrized, and contour-ordered correlators contain different information. The theorem states only that, once an ordering is fixed, its connected correlators are transduced linearly.

5.4 Commutator transduction

The commutator algebra is transferred as well.

For smeared observables,

$$[\widehat{T}(f), \widehat{T}(g)] = [\widehat{\tau}(L_e^\dagger f), \widehat{\tau}(L_e^\dagger g)]. \quad (217)$$

In local symbolic notation this becomes

$$[\widehat{T}(x), \widehat{T}(y)] = L_x [\widehat{\tau}(x), \widehat{\tau}(y)] L_y^T, \quad (218)$$

with the appropriate contractions over source and torsion indices.

Proposition 5.4 (Commutator transduction). *A nonvanishing commutator of a source-induced geometric observable can be inherited entirely from the noncommutativity of its quantum source:*

$$[\widehat{\tau}, \widehat{\tau}] \neq 0 \implies [\widehat{T}, \widehat{T}] = L[\widehat{\tau}, \widehat{\tau}]L^T, \quad (219)$$

provided the corresponding response directions are not annihilated by L .

Consequently,

$$[\widehat{T}(x), \widehat{T}(y)] \neq 0 \quad (220)$$

does not by itself establish an independently quantized gravitational field. It may arise entirely because the geometric observable is a noncommutative image of quantum matter.

5.5 Vanishing mean and nonvanishing geometric correlations

The distinction becomes especially clear when the source has zero mean.

Corollary 5.5 (Zero mean does not imply absence of quantum response). *Suppose*

$$\langle \widehat{\tau}(x) \rangle = 0 \quad (221)$$

while, for some pair of smeared or local observables,

$$\langle O \widehat{\tau}(x) \widehat{\tau}(y) \rangle_c \neq 0. \quad (222)$$

Then

$$\langle \widehat{T}(x) \rangle = 0, \quad (223)$$

but, provided the relevant correlated directions survive under L ,

$$\langle O \widehat{T}(x) \widehat{T}(y) \rangle_c \neq 0. \quad (224)$$

Thus

$$\langle \widehat{T} \rangle = 0 \not\Rightarrow \text{absence of quantum torsional response.} \quad (225)$$

The same conclusion applies to higher connected functions. A source state may possess vanishing one-point spin density while retaining higher connected source correlations, all of which are inherited by the source-induced geometric sector.

For a Gaussian or quasi-free source state, relative to the chosen correlation ordering, the connected hierarchy is determined by the two-point data. For a non-Gaussian state, higher connected torsion correlators retain information not contained in the covariance alone.

5.6 Spinor sources

For spinor matter, a particularly important source is the axial current

$$J_5^a = \bar{\psi} \gamma^a \gamma_5 \psi. \quad (226)$$

In Einstein–Cartan-type theories, the spin source entering the algebraic connection equation is related to this axial current, with coefficients depending on convention and on the precise matter normalization.

Schematically,

$$T \sim \kappa \epsilon \cdot J_5 \quad (227)$$

in the pure Einstein–Cartan channel, while a finite identity/Holst component can modify the tensor structure of the response.

Consequently, a quantum source correlator such as

$$\langle \widehat{J}_5(x) \widehat{J}_5(y) \rangle_c \quad (228)$$

induces the corresponding torsion correlation through Eq. (216).

Entangled or otherwise correlated matter states can therefore produce correlated geometric response even when the mean spin current vanishes.

The spatial range of this effect should not be overstated. If the torsion equation is algebraic, L_e is local. Correlations of torsional observables at separated points then reflect correlations already present in the source:

$$\text{nonlocal source correlation} \implies \text{nonlocal correlation of local geometric responses.} \quad (229)$$

This is not equivalent to an independently propagating long-range torsion field.

5.7 Comparison-loop readout

The source-induced torsional response may be probed operationally through a sufficiently small comparison loop.

Let

$$\gamma = \partial \Sigma$$

be a small closed loop bounding an oriented surface Σ , and let P_a be a probe covector in the relevant translational representation. To leading nontrivial order in the loop size, the scalar torsional closure defect has the schematic form

$$X_\Sigma(P) = \int_\Sigma P_a T^a. \quad (230)$$

At the quantum level define the smeared geometric loop observable

$$\widehat{X}_\Sigma(P) = \int_\Sigma P_a \widehat{T}^a. \quad (231)$$

Using $\widehat{T} = L_e \widehat{\tau}$,

$$\widehat{X}_\Sigma(P) = \int_\Sigma P_a L^a_A \widehat{\tau}^A, \quad (232)$$

with the obvious integral-kernel generalization when the response is nonlocal.

The corresponding leading comparison unitary is written schematically as

$$\widehat{W}_\gamma(P) = \exp \left[-\frac{i}{\hbar} \widehat{X}_\Sigma(P) \right] + O(\ell^3), \quad (233)$$

where ℓ denotes the characteristic loop size.

The correction term represents curvature contributions, source variation, path ordering, and higher terms in the small-loop expansion. Thus Eq. (233) is a leading local response formula and not an exact non-Abelian Stokes identity.

For the single smeared observable $\widehat{X}_\Sigma(P)$, define the characteristic comparison function

$$\chi_\Sigma(\lambda; P) := \left\langle \exp \left[-\frac{i\lambda}{\hbar} \widehat{X}_\Sigma(P) \right] \right\rangle. \quad (234)$$

Because this characteristic function involves powers of one and the same smeared operator, no ambiguity between different operator orderings arises within its ordinary one-variable cumulant expansion.

Whenever the characteristic function is sufficiently regular in a neighborhood of $\lambda = 0$,

$$\log \chi_\Sigma(\lambda; P) = \sum_{n=1}^{\infty} \frac{1}{n!} \left(-\frac{i\lambda}{\hbar} \right)^n \left\langle \widehat{X}_\Sigma(P)^n \right\rangle_c. \quad (235)$$

The source-transduction theorem therefore transfers directly into the statistics of the loop observable.

Theorem 5.6 (Loop readout of source-induced geometric cumulants). *In the linearized small-loop regime,*

$$\begin{aligned} & \left\langle \widehat{X}_\Sigma(P)^n \right\rangle_c \\ &= \int_{\Sigma^n} \prod_{r=1}^n [P_{a_r}(x_r) L^{a_r}_{A_r}(x_r)] \left\langle \widehat{\tau}^{A_1}(x_1) \cdots \widehat{\tau}^{A_n}(x_n) \right\rangle_c^{(\Sigma)}, \end{aligned} \quad (236)$$

where the superscript (Σ) denotes the ordering induced by the single smeared loop observable in the expansion of Eq. (234).

Thus a coherent comparison-loop experiment is, in principle, sensitive to the source-induced geometric cumulant hierarchy.

The theorem does not claim that every source correlator can be reconstructed independently from one experimental protocol. It states that in the purely slaved linear-response regime, the loop observable contains no independent quantum statistical information beyond that already present in the source and the response kernel.

If

$$\langle \widehat{\tau} \rangle = 0,$$

then

$$\langle \widehat{X}_\Sigma(P) \rangle = 0. \quad (237)$$

Nevertheless,

$$\left\langle \widehat{X}_\Sigma(P)^2 \right\rangle_c \neq 0 \quad (238)$$

whenever the integrated source covariance survives the response and smearing maps.

Thus a comparison loop can have zero mean closure defect while retaining nontrivial quantum variance and higher cumulants.

5.8 Hierarchy of geometric quantumness

It remains essential to separate different notions of quantum geometry.

Definition 5.7 (Geometric quantum-response hierarchy). *We distinguish the following four levels:*

1. Classical reconstructed geometry: e, ω, T, R are classical geometric response fields.
2. Quantum matter on classical geometry: matter observables are quantum, while geometry is evaluated only from classical expectation-value sources.
3. Source-induced operator geometry: geometric observables such as $\widehat{T}$ are operator valued because they are deterministic response images of quantum source operators.
4. Intrinsic quantum geometry: the gravitational sector contains independent operator degrees of freedom and quantum information not reducible to the matter-source algebra and a fixed response map.

The third level is genuinely operator valued but should not be confused with the fourth.

At the level of connected two-point functions, a more general theory may be decomposed schematically as

$$C_{TT}^{\text{full}} = LC_{\tau\tau}L^T + C_{TT}^{\text{intrinsic}} + C_{TT}^{\text{mixed}}. \quad (239)$$

The first term is the purely source-induced contribution. The second denotes correlations associated with independent gravitational quantum degrees of freedom, while the third denotes matter–geometry correlations that cannot be assigned entirely to either sector.

Equation (239) is schematic unless a microscopic operator-algebraic or Hilbert-space decomposition has been specified. It should therefore be understood as an effective diagnostic rather than as a universal canonical decomposition.

Proposition 5.8 (Criterion for purely slaved geometric quantumness). *If every geometric correlation function in an effective sector can be expressed entirely as the image of matter-source correlation functions through a fixed response map L , then the quantum information in that geometric sector is source induced on that effective description.*

Conversely, geometric quantum correlation data that remain after the source-induced and mixed contributions have been separated require additional geometric quantum information beyond the fixed source map.

The same diagnostic applies to commutators. In a purely slaved sector,

$$[\widehat{T}, \widehat{T}] = L[\widehat{\tau}, \widehat{\tau}]L^T. \quad (240)$$

An additional geometric commutator contribution that cannot be represented in this form would signal independent quantum structure in the effective gravitational sector.

5.9 Algebraic response feedback

The same algebraic torsion channel that converts a spin source into a geometric response also generates an effective interaction among matter degrees of freedom when the torsion variable is eliminated.

Consider a schematic first-order action in which the relevant torsional variable has no independent kinetic term and appears quadratically:

$$S = S_0[e, \omega_{\text{tf}}] + S_m + \frac{1}{2} \int T K_T T + \int T \tau. \quad (241)$$

Here ω_{tf} denotes the torsion-free or remaining connection variables, and K_T is an invertible local torsional stiffness operator on the channel under consideration.

The overall signs and numerical normalizations depend on convention.
Variation with respect to T gives

$$K_T T + \tau = 0, \quad (242)$$

hence

$$T = -K_T^{-1} \tau. \quad (243)$$

Substitution back into the action yields

$$S_{\text{eff}} = S_0 + S_m - \frac{1}{2} \int \tau K_T^{-1} \tau. \quad (244)$$

Theorem 5.9 (Algebraic response-feedback theorem). *If a geometric response variable enters an effective action quadratically and algebraically and is linearly coupled to a matter source, then eliminating that geometric variable produces an effective bilinear source interaction,*

$$S_{\text{int}}^{\text{eff}} = -\frac{1}{2} \int \tau K_T^{-1} \tau, \quad (245)$$

with the same inverse stiffness controlling the algebraic source-response channel.

The reciprocal structure is therefore

$$\text{source} \longrightarrow \text{geometric response} \longrightarrow \text{effective source interaction}. \quad (246)$$

For Dirac matter, the relevant spin source is typically related to the axial current. The induced interaction then takes the schematic local four-fermion form

$$S_{\text{eff}}^{(4\psi)} \sim G_T \int d^4x J_{5a} J_5^a, \quad (247)$$

where G_T is proportional to the appropriate torsional compliance. Its precise coefficient depends on the normalization and on the constitutive channel, including possible Holst-type contributions.

In a suitable nonrelativistic low-energy sector of two overlapping spin- $\frac{1}{2}$ systems, the spin-dependent part may reduce schematically to

$$H_T = J_T \mathbf{S}_A \cdot \mathbf{S}_B \quad (248)$$

plus state-independent or additional channel-dependent terms.

Such an interaction can generate entanglement from suitable initially separable spin states. For example, because $\mathbf{S}_A \cdot \mathbf{S}_B$ has different eigenvalues in the singlet and triplet sectors, generic superpositions of these sectors acquire different phases under time evolution.

The conditional response chain is therefore

$$\text{quantum spin state} \longrightarrow \text{torsional response} \longrightarrow \text{effective spin interaction}. \quad (249)$$

This interaction should not be interpreted as a propagating long-range torsion force when the torsion equation is algebraic. Eliminating an auxiliary or algebraic geometric variable produces a contact or overlap interaction without introducing an independent asymptotic torsion quantum.

Thus

$$\text{torsion-induced entangling interaction} \not\Rightarrow \text{propagating quantum torsion}. \quad (250)$$

Likewise, a nonzero connected geometric correlation

$$\langle \widehat{T}(x) \widehat{T}(y) \rangle_c \neq 0 \quad (251)$$

is not by itself a Bell-nonlocality criterion. A Bell violation additionally requires an appropriate entangled state, suitable local incompatible observables, and the corresponding measurement statistics.

5.10 Generating functionals and operator ordering

The source-transduction structure admits a compact generating-functional formulation, but the operator-ordering convention must be specified carefully.

First consider the ordinary quantum characteristic functional

$$Z_\tau^W[J] := \left\langle \exp \left[\frac{i}{\hbar} \widehat{\tau}(J) \right] \right\rangle, \quad \widehat{\tau}(J) := \int J \cdot \widehat{\tau}, \quad (252)$$

whenever the exponential is well defined.

The superscript W emphasizes that functional differentiation of an ordinary exponential of a non-commuting operator-valued field generates the corresponding Weyl/symmetrically ordered multilinear moments, rather than an arbitrary fixed Wightman or time ordering.

For the source-induced torsion field,

$$\widehat{T}(K) = \widehat{\tau}(L_e^\dagger K),$$

and therefore the characteristic functionals satisfy the exact identity

$$Z_T^W[K] = Z_\tau^W[L_e^\dagger K]. \quad (253)$$

Define

$$W_\tau^W[J] := -i\hbar \log Z_\tau^W[J], \quad W_T^W[K] := -i\hbar \log Z_T^W[K]. \quad (254)$$

Then

$$W_T^W[K] = W_\tau^W[L_e^\dagger K]. \quad (255)$$

Corollary 5.10 (Symmetrized generating-functional transduction). *In the purely linear source-induced sector, the full hierarchy of Weyl/symmetrically ordered connected cumulants of the torsional response is the pullback of the corresponding source cumulants under $L_e^\dagger$:*

$$W_T^W = W_\tau^W \circ L_e^\dagger. \quad (256)$$

This statement should not be confused with the generation of arbitrary operator orderings.

For example, time-ordered connected correlators are generated instead from

$$Z_\tau^T[J] = \left\langle \mathcal{T} \exp \left[\frac{i}{\hbar} \int J \cdot \widehat{\tau} \right] \right\rangle. \quad (257)$$

For an instantaneous local c-number response map that does not alter the time-ordering labels,

$$Z_T^T[K] = Z_\tau^T[L_e^\dagger K]. \quad (258)$$

Other Wightman or real-time orderings require the corresponding multi-source or Schwinger–Keldysh generating functional.

If the response kernel is genuinely nonlocal in time, the ordering must be defined together with the convolution structure; one should not replace it automatically by the simple local identity Eq. (258).

This distinction resolves the operator-ordering ambiguity while preserving the central source-transduction result.

The generating-functional formulation also gives a sharp diagnostic for new gravitational quantum information. If, after the appropriate ordering prescription, mixed terms, and renormalizations have been accounted for, the complete geometric generating functional cannot be represented as a pullback of the source functional through the gravitational response map, then the geometric sector contains quantum information not inherited from the source alone.

5.11 Operational constitutive phase from source response

The source-to-geometry response also depends on the constitutive orientation. Since

$$L_e = \mathcal{T}_e^{-1} C^{-1} \quad (259)$$

and

$$C^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}, \quad (260)$$

the complete source-induced correlation hierarchy depends on both the identity and Hodge constitutive channels.

Consequently, two theories whose spinless classical vacuum equations are insensitive to a constitutive parameter can nevertheless produce different spin–torsion response tensors, different loop statistics, or different effective spin interactions.

This suggests defining the constitutive orientation operationally from the measured source-to-torsion susceptibility rather than identifying it directly with a bare action coefficient.

Let

$$\chi_{T\tau}$$

denote the appropriate low-frequency spin-to-torsion susceptibility. After the fixed solder-dependent tensor maps have been removed, suppose the two Lorentz-equivariant constitutive channels can be resolved as

$$\chi_{T\tau} = \chi_I I + \chi_\star \star. \quad (261)$$

When χ_I and $\chi_\star$ are real in the local nondissipative limit and are not simultaneously zero, define

$$\theta_{\text{op}} = \text{atan2}(\chi_\star, \chi_I) \pmod{2\pi}. \quad (262)$$

Equivalently, where $\chi_I \neq 0$,

$$\tan \theta_{\text{op}} = \frac{\chi_\star}{\chi_I}. \quad (263)$$

In a general dynamical or dissipative response problem the coefficients may become complex and frequency dependent. In that case a single real angle need not be sufficient; the operational phase must be defined from the full retarded response, as discussed in the next section.

This operational definition is preferable when field redefinitions, boundary terms, or Holst–Nieh–Yan rearrangements render a bare action parameter scheme dependent. What is directly physical is the measurable source-to-response kernel.

5.12 Summary of the source-induced quantum sector

The main result of this section may be summarized as

$$\widehat{\tau} \longrightarrow \widehat{T} = L_e \widehat{\tau}, \quad (264)$$

$$\langle O \widehat{\tau}^n \rangle_c \longrightarrow \langle O \widehat{T}^n \rangle_c, \quad (265)$$

$$[\widehat{\tau}, \widehat{\tau}] \longrightarrow [\widehat{T}, \widehat{T}], \quad (266)$$

$$\widehat{T} \longrightarrow \widehat{X}_\Sigma(P) \longrightarrow \text{comparison-loop statistics}, \quad (267)$$

and

$$K_T^{-1} \longrightarrow \text{effective matter-matter interaction.} \quad (268)$$

The first bridge developed in the preceding sections was

$$\text{spin-cover complex structure} \longrightarrow \text{gravitational Hodge structure.} \quad (269)$$

The present section establishes a second, logically distinct bridge:

$$\text{quantum matter source} \longrightarrow \text{operator-valued geometric response.} \quad (270)$$

The limitation is equally important. Every geometric quantum correlation derived in the present section is slaved to the matter source through the fixed response map L_e . No independent gravitational oscillator, intrinsic gravitational vacuum fluctuation, or autonomous gravitational operator algebra has been required.

Thus

$$\text{source-induced operator geometry} \not\Rightarrow \text{intrinsic quantum gravity.} \quad (271)$$

The next problem is therefore not whether the source-induced torsion observable can be operator valued; it can. The next problem is whether the gravitational response coefficient itself can be characterized operationally and dynamically.

In particular, the constitutive stiffness K_G should be related to the inverse source-to-geometry susceptibility, while a full quantum response description must distinguish causal response from quantum noise. The next section develops these issues using retarded response functions, spectral representations, and the Kubo framework.

6 Gravitational Stiffness, Kubo Response, and Quantum Noise

The preceding sections identify two conceptually distinct structures in the gravitational response.

First, Lorentz covariance together with the comparison-constitutive compatibility condition can constrain the *direction* of the constitutive map. In the pure Hodge branch,

$$C = \varepsilon_C K_G \star, \quad \varepsilon_C = \pm 1, \quad K_G > 0. \quad (272)$$

Second, the source-transduction relation shows that the inverse magnitude K_G^{-1} controls the amplitude of the algebraic torsional response to spin sources. The remaining question is whether K_G can be assigned a direct operational meaning independently of its appearance as a coefficient in a chosen gravitational action.

At the level of response theory, the answer is affirmative. After fixed tensorial and differential structures of a normalized response channel have been factored out, the coefficient multiplying the inverse response kernel is naturally interpreted as a stiffness, while its inverse is a compliance.

This interpretation does not determine the numerical value of the gravitational stiffness from kinematics alone.

6.1 Gravitational compliance

For the pure Hodge constitutive map Eq. (272), the algebraic spin–torsion response derived in Section 5 is

$$T = -\varepsilon_C K_G^{-1} \mathcal{T}_e^{-1} \star \tau. \quad (273)$$

After the fixed tensor map

$$-\varepsilon_C \mathcal{T}_e^{-1} \star$$

has been factored out, the response amplitude is proportional to K_G^{-1} .

This motivates the following terminology.

Definition 6.1 (Gravitational compliance and stiffness). *Consider a normalized gravitational response channel in which a generalized source J produces a geometric deformation G according to*

$$G = \chi_G J. \quad (274)$$

After fixed kinematical tensor structures and conventional normalization factors have been removed, the coefficient χ_G will be called the gravitational compliance of that channel. Its inverse coefficient, when defined, will be called the corresponding gravitational stiffness.

In the algebraic torsion channel,

$$\chi_T \propto K_G^{-1}. \quad (275)$$

A larger K_G therefore corresponds to a smaller geometric response to a fixed normalized spin source, whereas a smaller K_G corresponds to a more compliant channel.

This interpretation extends to propagating response, but the scalar language must then be used carefully. A dynamical field has an inverse response kernel containing differential operators, projectors, and possibly nonlocal form factors. The gravitational stiffness is therefore generally the coefficient of a normalized infrared inverse-response operator rather than literally the reciprocal of a constant zero-momentum susceptibility.

6.2 Quadratic effective action and inverse response

Let

$$\mathcal{A}^A(x)$$

denote a collection of linearized geometric response variables and $J_A(x)$ their external sources. The collective index A may contain spacetime, Lorentz, polarization, or channel labels.

Suppose the quadratic effective action around a specified background takes the form

$$\Gamma^{(2)}[\mathcal{A}; J] = \frac{1}{2} \int d^4x d^4y \mathcal{A}^A(x) \Gamma_{AB}^{(2)}(x, y) \mathcal{A}^B(y) - \int d^4x J_A(x) \mathcal{A}^A(x). \quad (276)$$

The linearized field equation is

$$\int d^4y \Gamma_{AB}^{(2)}(x, y) \mathcal{A}^B(y) = J_A(x). \quad (277)$$

Whenever the inverse exists on the physical response sector,

$$\mathcal{A}^A(x) = \int d^4y K^{AB}(x, y) J_B(y), \quad (278)$$

where, schematically,

$$K = \left(\Gamma^{(2)} \right)^{-1}. \quad (279)$$

For a gauge theory this inverse must not be interpreted on the unreduced field space without qualification. Gauge directions generate zero modes or redundancies in the quadratic kernel. One must first impose a gauge fixing, project to gauge-invariant physical variables, or formulate the inverse on the reduced phase space using the appropriate Dirac or reduced kernel.

Suppose that, after this reduction and after a normalization of the selected response channel, the retarded inverse response has the infrared form

$$\Gamma_R^{(2)}(k) = K_G D_R(k) + \Gamma_{\text{sub}}(k), \quad (280)$$

where $D_R(k)$ contains the universal tensorial and derivative structure of the channel and Γ_{sub} contains mixing, higher-derivative, analytic, or otherwise subleading contributions.

When the first term dominates and D_R is invertible on the reduced physical sector,

$$K_R(k) \simeq K_G^{-1} D_R^{-1}(k). \quad (281)$$

Proposition 6.2 (Stiffness as an inverse response coefficient). *Suppose a reduced and normalized gravitational channel has infrared inverse retarded response*

$$\Gamma_R^{(2)} = K_G D_R \quad (282)$$

to leading order. Then its source-to-geometry retarded response is

$$K_R = K_G^{-1} D_R^{-1} \quad (283)$$

on that channel. Thus K_G is the stiffness coefficient and K_G^{-1} the associated compliance coefficient.

This proposition does not assert that the full gravitational susceptibility is momentum independent. A propagating massless channel contains poles and tensor projectors. The constant K_G is the normalization multiplying the universal low-energy inverse kernel.

For Einstein–Hilbert normalization one has schematically

$$K_G \propto \frac{c^3}{G}, \quad (284)$$

up to conventional numerical factors such as 8π or 16π . Consequently,

$$K_G^{-1} \propto \frac{G}{c^3}. \quad (285)$$

In this response sense, a larger Newton coupling corresponds to a larger gravitational compliance.

6.3 Retarded Kubo response

The causal quantum version of a susceptibility is the retarded response kernel.

Let

$$\widehat{O}_A(x)$$

be Hermitian quantum observables conjugate to generalized external sources $J_A(x)$. With the source-sign convention used here, define

$$K_{AB}^R(x, y) := -\frac{i}{\hbar} \theta(x^0 - y^0) \langle [\widehat{O}_A(x), \widehat{O}_B(y)] \rangle. \quad (286)$$

If the interaction with the source is written with the opposite sign, the overall sign convention for K_R changes correspondingly.

For a weak perturbing source $\delta J_B(y)$, linear response gives

$$\delta \langle \widehat{O}_A(x) \rangle = \int d^4y K_{AB}^R(x, y) \delta J_B(y). \quad (287)$$

Equation (286) provides an operational quantum definition of gravitational susceptibility whenever the corresponding physical geometric observable and its conjugate source have been identified.

For the source-induced torsion sector, one may study the response of torsional observables to the matter spin source. More generally, one may consider metric, soldering, connection, curvature, or comparison-loop observables, provided their source coupling and gauge reduction are specified consistently.

It is useful to distinguish three concepts:

$$\text{static response,} \quad \text{dissipative response,} \quad \text{quantum fluctuation.} \quad (288)$$

The retarded susceptibility contains the first two. Its dispersive part controls the reactive response, while its absorptive part records dissipation when a dissipative channel exists.

Quantum fluctuation is instead encoded in the symmetrized covariance kernel.

6.4 Quantum noise

Define

$$\Delta \widehat{O}_A(x) = \widehat{O}_A(x) - \langle \widehat{O}_A(x) \rangle \quad (289)$$

and

$$N_{AB}(x, y) := \frac{1}{2} \langle \{ \Delta \widehat{O}_A(x), \Delta \widehat{O}_B(y) \} \rangle. \quad (290)$$

We shall refer to N_{AB} as the *quantum noise kernel*.

The terminology does not imply that the underlying physics is a classical stochastic process. Equation (290) is an ordinary symmetrized quantum correlation function. A stochastic-noise interpretation requires an additional effective or semiclassical construction.

6.5 Spectral representation and translation invariance

A Fourier representation in both frequency and spatial momentum requires more than stationarity.

Assume first that the state and background are invariant under time translations. Then two-point kernels depend on the time difference,

$$K(x^0, y^0; \mathbf{x}, \mathbf{y}) = K(x^0 - y^0; \mathbf{x}, \mathbf{y}),$$

and one may Fourier transform with respect to the time difference:

$$K(\omega; \mathbf{x}, \mathbf{y}).$$

To use a diagonal momentum-space representation

$$K(\omega, \mathbf{k}),$$

we additionally assume spatial translation invariance, so that

$$K(x, y) = K(x - y). \quad (291)$$

Under this stationarity and spatial homogeneity assumption, define the commutator spectral density

$$\rho_{AB}(\omega, \mathbf{k}) := \int d^4x e^{i\omega t - i\mathbf{k} \cdot \mathbf{x}} \langle [\widehat{O}_A(x), \widehat{O}_B(0)] \rangle. \quad (292)$$

Similarly, define the noise spectrum

$$N_{AB}(\omega, \mathbf{k}) := \frac{1}{2} \int d^4x e^{i\omega t - i\mathbf{k} \cdot \mathbf{x}} \langle \{ \Delta \widehat{O}_A(x), \Delta \widehat{O}_B(0) \} \rangle. \quad (293)$$

If spatial translation invariance is absent, one should instead retain objects such as

$$\rho_{AB}(\omega; \mathbf{x}, \mathbf{y}), \quad N_{AB}(\omega; \mathbf{x}, \mathbf{y}), \quad (294)$$

or use an appropriate Wigner transform. A single momentum $\mathbf{k}$ then does not label the complete two-point response.

With the Fourier and Kubo conventions above, the retarded kernel obeys the usual causal spectral representation. In a diagonal Hermitian channel,

$$K_R(\omega, \mathbf{k}) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} d\omega' \frac{\rho(\omega', \mathbf{k})}{\omega - \omega' + i0^+}, \quad (295)$$

up to the overall sign associated with the source convention.

For the convention of Eq. (286), the corresponding absorptive part satisfies

$$\text{Im } K_R(\omega, \mathbf{k}) = -\frac{1}{2\hbar} \rho(\omega, \mathbf{k}) \quad (296)$$

for the diagonal Hermitian channel.

6.6 KMS condition and fluctuation–dissipation relation

Suppose now that the stationary state is a thermal KMS state at inverse temperature

$$\beta = \frac{1}{k_B T}.$$

For a homogeneous equilibrium state, the KMS relation connects the positive- and negative-frequency correlation functions and yields the fluctuation–dissipation theorem.

For a diagonal Hermitian response channel,

$$N(\omega, \mathbf{k}) = \frac{1}{2} \coth\left(\frac{\beta\hbar\omega}{2}\right) \rho(\omega, \mathbf{k}). \quad (297)$$

Using Eq. (296), the same relation can be written, with the present retarded-sign convention, as

$$N(\omega, \mathbf{k}) = -\hbar \coth\left(\frac{\beta\hbar\omega}{2}\right) \text{Im } K_R(\omega, \mathbf{k}). \quad (298)$$

If the opposite sign convention is adopted for the source coupling and retarded susceptibility, the corresponding sign in Eq. (298) changes. The invariant content is the relation between equilibrium symmetrized fluctuations and the dissipative spectral response.

For several coupled Hermitian channels, the KMS condition gives the corresponding matrix-valued relation, with index order, transpose, and complex-conjugation conventions handled consistently.

Theorem 6.3 (Equilibrium fluctuation–response relation). *Let the quantum state be a KMS state with respect to the stationary Hamiltonian evolution. For a spatially homogeneous equilibrium background, the symmetrized noise spectrum of a Hermitian geometric response channel is fixed by the commutator spectral density through Eq. (297). The same commutator spectral density determines the absorptive part of the retarded susceptibility.*

Thus equilibrium geometric fluctuations and causal dissipative response are not independent data.

The spatial homogeneity assumption in the momentum-space statement is not essential to the physical content of the fluctuation–dissipation theorem. Without it, the theorem should instead be written in the mixed frequency–position representation.

6.7 Inherited response and noise in the source-induced sector

The fluctuation–response structure becomes particularly transparent for source-induced geometry. Suppose

$$\widehat{T} = L\widehat{\tau} \quad (299)$$

with a fixed state-independent real local response map L . Then

$$N_{TT} = L N_{\tau\tau} L^T \quad (300)$$

and

$$K_{TT}^R = L K_{\tau\tau}^R L^T. \quad (301)$$

For a translation-invariant but momentum-dependent response kernel, the Fourier-space relation should be written more carefully as

$$N_{TT}(k) = L(k) N_{\tau\tau}(k) L(-k)^T, \quad (302)$$

and similarly for the commutator or retarded correlator whenever the same linear convolution map acts on both operator insertions.

For a real kernel,

$$L(-k)^T$$

is equivalently represented by the appropriate Hermitian adjoint under the standard Fourier-space inner-product convention.

Hence a fluctuation–dissipation relation satisfied by the source is automatically inherited by the purely source-induced geometric channel, provided the same deterministic linear response map is used consistently.

This reinforces the distinction introduced in Section 5:

$$\text{geometric quantum noise} \not\Rightarrow \text{intrinsic gravitational quantum degrees of freedom.} \quad (303)$$

6.8 Operational constitutive orientation

The causal-response formulation also provides an operational way to describe the constitutive orientation.

After the fixed solder-dependent maps have been removed, suppose the retarded spin-to-torsion response can be decomposed into the two local Lorentz-equivariant bivector channels,

$$K_{T\tau}^R(\omega, \mathbf{k}) = \chi_I(\omega, \mathbf{k})I + \chi_\star(\omega, \mathbf{k}) \star. \quad (304)$$

In a nondissipative local regime in which χ_I and $\chi_\star$ may be taken as real coefficients after a common normalization has been removed, define

$$\theta_{\text{op}}(\omega, \mathbf{k}) = \text{atan2}(\chi_\star(\omega, \mathbf{k}), \chi_I(\omega, \mathbf{k})). \quad (305)$$

Equivalently,

$$\tan \theta_{\text{op}} = \frac{\chi_\star}{\chi_I} \quad (306)$$

where the denominator is nonzero.

In a dissipative or genuinely nonlocal regime, χ_I and $\chi_\star$ are generally complex functions. In that case a single real angle is insufficient to characterize the response. A more invariant quantity is the complex channel ratio

$$r_{\star/I}(\omega, \mathbf{k}) := \frac{\chi_\star(\omega, \mathbf{k})}{\chi_I(\omega, \mathbf{k})}, \quad (307)$$

where defined, together with the overall response normalization.

Only when this ratio is real, or when the two channels share a removable common complex phase, does it reduce to a single real constitutive angle.

Thus a bare parameter such as a Holst or Immirzi coefficient need not be the most direct physical observable. The measurable object is the source-to-geometry response tensor itself.

6.9 Running and complex gravitational stiffness

The same qualification applies to the gravitational stiffness. In a general effective theory one should expect momentum- and frequency-dependent response:

$$K_G \longrightarrow K_G(\omega, \mathbf{k}). \quad (308)$$

In a weakly dissipative low-frequency channel, the real analytic part of the inverse response determines the elastic or static stiffness, while an imaginary part records absorptive response when such a channel is kinematically allowed.

More generally, one should distinguish the local renormalized coefficient multiplying the leading derivative operator from the complete inverse response function. The latter may contain poles, cuts, thresholds, nonlocality, and mixing.

Thus

$$K_G(\omega, \mathbf{k})$$

is best regarded as shorthand for a channel-normalized effective coefficient, not necessarily as a fundamental scalar coupling at arbitrary frequency and momentum.

6.10 Subtracted dispersion relations and Newton stiffness

Causality constrains the analytic structure of the retarded response but does not, by itself, determine every real local coefficient.

For a causal scalar response channel one may write a suitably subtracted dispersion relation schematically as

$$\operatorname{Re} K_R(\omega) = P_n(\omega) + \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\operatorname{Im} K_R(\omega')}{\omega' - \omega}, \quad (309)$$

where $P_n(\omega)$ denotes the subtraction polynomial required by the ultraviolet behavior and $\mathcal{P}$ denotes the principal value.

In a more general tensor or matrix channel the same statement holds componentwise or after projection onto appropriate physical response eigenchannels.

The subtraction data represent analytic local contributions that are not determined by the absorptive spectrum alone. In an effective action, local counterterms contribute precisely to this analytic part.

The Einstein–Hilbert coefficient is a local low-derivative gravitational coefficient. Its renormalized numerical value therefore cannot in general be reconstructed solely from the positive-frequency spectral weight of propagating matter or gravitational excitations.

Proposition 6.4 (Spectral underdetermination of Newton stiffness). *Causality, Kubo response theory, unitarity constraints on physical spectral densities, and dispersion relations constrain the analytic and absorptive structure of gravitational response. They do not, in general, determine the complete numerical low-energy value of the local gravitational stiffness K_G .*

Subtraction constants, local counterterms, renormalization conditions, and ultraviolet completion data can modify the real analytic part without being fixed by the absorptive spectrum alone.

The logical boundary is therefore

$$\text{response interpretation of } G \not\Rightarrow \text{numerical derivation of } G. \quad (310)$$

The MS reconstruction may constrain the physical role played by Newton’s coupling without predicting its measured renormalized value.

6.11 First-order presymplectic normalization

There is a second quantum interpretation of the same stiffness coefficient, arising from the first-order presymplectic structure.

For a pure Hodge gravitational constitutive term, the first-order action has a presymplectic potential of the schematic form

$$\Theta_{\text{grav}} = K_G \int_{\Sigma} (\star B)_{ab} \wedge \delta \omega^{ab} + \Theta_{\text{constraint}} + \Theta_{\text{boundary}}, \quad (311)$$

where

$$B = e \wedge e \quad (312)$$

and ω^{ab} is the Lorentz connection.

The corresponding presymplectic two-form contains

$$\Omega_{\text{grav}} = K_G \int_{\Sigma} \delta(\star B)_{ab} \wedge_F \delta \omega^{ab} + \dots, \quad (313)$$

where $\wedge_F$ denotes antisymmetrization in field space and the omitted terms include constraints and boundary contributions.

Before gauge reduction and imposition of gravitational constraints, Ω_{grav} is presymplectic rather than a nondegenerate physical symplectic form.

Suppose, however, that a reduced canonical channel is obtained for which the induced symplectic form retains the overall normalization K_G . Then the inverse reduced symplectic structure scales as

$$\{\omega, \star B\}_{\text{red}} \sim K_G^{-1}, \quad (314)$$

with tensorial delta functions and numerical normalization factors suppressed.

After quantization,

$$[\widehat{\omega}, \widehat{\star B}]_{\text{red}} \sim i\hbar K_G^{-1}. \quad (315)$$

The subscript “red” is essential. The exact physical bracket depends on the constraint system, gauge fixing, boundary conditions, choice of canonical variables, and resulting Dirac or reduced symplectic structure.

Equation (315) is therefore a scaling relation, not a universal unreduced tetrad-gravity commutator.

Proposition 6.5 (Quantum conjugacy scale of gravitational compliance). *Suppose a reduced first-order gravitational channel has symplectic normalization proportional to K_G . Then the characteristic quantum scale of noncommutativity between its normalized conjugate geometric variables is proportional to*

$$\hbar K_G^{-1}. \quad (316)$$

Using

$$K_G \propto \frac{c^3}{G}, \quad (317)$$

one obtains

$$\hbar K_G^{-1} \propto \frac{\hbar G}{c^3}. \quad (318)$$

The combination on the right-hand side has the dimensions of area and is the Planck area,

$$\ell_P^2 = \frac{\hbar G}{c^3}. \quad (319)$$

Hence

$$\ell_P^2 \sim \hbar \times \text{gravitational compliance}, \quad (320)$$

up to conventional normalization factors.

This is a response-theoretic interpretation of the Planck area. It is strictly weaker than the claim that physical spacetime possesses a literal minimum area.

Nothing in Eq. (320) implies that a physical area operator has a discrete spectrum, much less a smallest nonzero eigenvalue. The statement is only that the natural dimensionful quantum conjugacy scale obtained by multiplying the gravitational compliance by $\hbar$ is proportional to ℓ_P^2 .

6.12 Gauge-theory analogy

A structurally similar normalization occurs in ordinary Yang–Mills theory. With the schematic action

$$S_{\text{YM}} \sim \frac{1}{g^2} \int F \wedge \star F, \quad (321)$$

the normalization of the canonical momentum contains the inverse gauge coupling. After canonical normalization, one obtains schematically

$$[\widehat{A}, \widehat{E}] \sim i\hbar g^2, \quad (322)$$

when E denotes an excitation variable normalized so that the coupling is carried by the canonical pairing.

Both internal gauge theory and first-order gravity therefore exhibit the structural pattern

$$[\text{comparison coordinate}, \text{conjugate excitation}] \sim i\hbar \times \text{compliance}. \quad (323)$$

The analogy is structural only. The gravitational connection belongs to the Lorentz geometric sector, while an internal gauge connection acts on an independent internal sector. Their gauge groups, constraints, source representations, observables, and dynamics remain distinct.

6.13 Three meanings of gravitational stiffness

The same coefficient K_G can therefore appear in three related low-energy structures.

From static response,

$$K_G^{-1} \sim \text{static source-to-geometry compliance}. \quad (324)$$

From the retarded response,

$$K_G^{-1} \sim \text{normalization of the infrared causal response}. \quad (325)$$

From the reduced first-order symplectic structure,

$$\hbar K_G^{-1} \sim \text{quantum geometric conjugacy scale}. \quad (326)$$

If a single local low-energy effective action governs all three descriptions, one expects the same renormalized local coefficient to appear in each after normalization.

This motivates the consistency condition

$$K_G^{\text{static}} = K_G^{\text{retarded}} = K_G^{\text{symplectic}} \quad (327)$$

in the common local infrared limit, after all quantities have been defined in the same background, field normalization, renormalization scheme, and physical channel.

Equation (327) is not an identity that should be assumed at arbitrary frequency or scale. Nonlocality, field redefinitions, dissipative effects, mixing, running couplings, or distinct effective normalizations can cause the three operational definitions to differ away from the common infrared regime.

Agreement would therefore be a nontrivial consistency property of the effective gravitational theory.

6.14 Noise is not susceptibility

The causal-response description sharpens the distinction between gravitational stiffness and geometric fluctuation.

A large noise kernel

$$N_{GG}$$

does not by itself imply a large gravitational compliance. The fluctuation amplitude depends on the state as well as on the response operator.

Conversely, a highly compliant channel need not display large fluctuations in an arbitrary nonequilibrium or low-occupation state.

Only under additional equilibrium conditions, such as the KMS hypothesis, does the fluctuation–dissipation theorem relate noise to absorptive response.

Thus,

$$\text{geometric fluctuation amplitude} \not\equiv \text{gravitational susceptibility}. \quad (328)$$

Likewise,

$$\text{geometric quantum noise} \not\Rightarrow \text{intrinsic quantum gravity}. \quad (329)$$

In a source-induced sector, Eq. (300) shows explicitly that geometric noise can be inherited entirely from the quantum source.

For a more general theory, one may write schematically

$$N_{GG}^{\text{full}} = LN_{JJ}L^\dagger + N_{GG}^{\text{intrinsic}} + N_{GG}^{\text{mixed}}, \quad (330)$$

where the precise meaning of the three terms depends on the microscopic operator basis and on how matter and gravitational degrees of freedom are separated.

Similarly,

$$K_{R,GG}^{\text{full}} = K_{R,GG}^{\text{induced}} + K_{R,GG}^{\text{intrinsic}} + K_{R,GG}^{\text{mixed}}. \quad (331)$$

These are effective diagnostic decompositions rather than universal canonical splittings.

A genuinely independent quantum gravitational sector should contribute response or fluctuation structures that cannot be reduced to the matter-source algebra and the fixed source-transduction kernel alone.

6.15 Effective scale dependence

The response viewpoint also makes scale dependence transparent.

Let

$$\mu$$

denote a coarse-graining or renormalization scale. If the effective local constitutive response remains closed inside the Lorentz-equivariant algebraic bivector channel, then it can be written as

$$C_{\text{eff}}(\mu) = \alpha(\mu)I + \beta(\mu)\star = K(\mu)e^{\theta(\mu)\star}. \quad (332)$$

Both the stiffness and the constitutive orientation may therefore run.

If the pure Hodge or Einstein constitutive direction is an infrared attractor, one would require

$$\theta(\mu) \longrightarrow \varepsilon_C \frac{\pi}{2} \quad \text{as } \mu \rightarrow 0, \quad (333)$$

together with a nonvanishing finite infrared stiffness,

$$K(\mu) \longrightarrow K_G > 0. \quad (334)$$

The second condition is important: convergence of an unnormalized constitutive operator toward zero cannot by itself determine its direction in the constitutive plane.

Neither Eq. (333) nor Eq. (334) follows from the local Lorentz representation theory. They are dynamical infrared hypotheses.

Moreover, even the parametrization Eq. (332) requires an effective-channel closure assumption. Coarse graining can generate higher derivatives, nonlocal form factors, additional torsional invariants, parity-sensitive structures, or mixing with other fields and response channels.

Thus

$$C_{\text{eff}}(\mu) \in \mathbb{R}I \oplus \mathbb{R}\star \quad (335)$$

is not guaranteed solely by the classification of local algebraic Lorentz-equivariant endomorphisms.

The dynamical closure of this channel, together with mixing between gravitational and other comparison sectors, is the subject of the next section.

6.16 Summary

The main relations established in this section are

$$K_G^{-1} \longrightarrow \text{gravitational compliance}, \quad (336)$$

$$K_R \longrightarrow \text{causal quantum response}, \quad (337)$$

$$N \longrightarrow \text{symmetrized quantum geometric noise}, \quad (338)$$

$$\text{KMS equilibrium} \longrightarrow \text{fluctuation-dissipation relation}, \quad (339)$$

and

$$\hbar K_G^{-1} \longrightarrow \text{quantum geometric conjugacy scale}. \quad (340)$$

With

$$K_G \propto \frac{c^3}{G},$$

the latter becomes

$$\hbar K_G^{-1} \propto \frac{\hbar G}{c^3} = \ell_P^2. \quad (341)$$

The structural interpretation is therefore stronger than the statement that Newton's constant is merely a coefficient inserted into an action. Its inverse stiffness controls the normalization of low-energy gravitational compliance and, in an appropriate reduced first-order canonical channel, the scale of quantum geometric conjugacy.

The numerical value of that stiffness remains dynamical and renormalized. Neither Lorentz representation theory, complex-comparison rigidity, Kubo response theory, spectral positivity, nor the fluctuation-dissipation theorem fixes it by itself.

7 Effective Response Flow, Einstein-Channel Closure, and Coupled Comparison Sectors

The preceding sections have identified three distinct roles of the gravitational constitutive response.

First, four-dimensional Lorentz covariance restricts a local algebraic constitutive map on the Lorentz bivector carrier to

$$C = \alpha I + \beta \star. \quad (342)$$

Second, the comparison–constitutive compatibility condition developed in Section 4 selects the pure Hodge direction

$$C = \varepsilon_C K_G \star, \quad \varepsilon_C = \pm 1. \quad (343)$$

Third, the magnitude K_G controls both the inverse source-to-geometry response and, in an appropriate reduced first-order description, the normalization of the gravitational symplectic structure.

These statements are local and infrared in character. A microscopic theory must additionally explain whether the same constitutive organization survives coarse graining.

This is not automatic. Integrating out short-distance degrees of freedom can generate higher-derivative invariants, nonlocal form factors, torsional operators, parity-sensitive structures, and mixing with additional comparison sectors. The pointwise representation-theoretic identity

$$\text{End}_{SO^+(1,3)}(\Lambda^2 V) = \mathbb{R}I \oplus \mathbb{R}\star$$

classifies local algebraic equivariant endomorphisms of the bivector carrier; it does not classify the complete quantum effective action.

We therefore distinguish exact local constitutive classification from dynamical closure of the low-energy response.

7.1 Effective constitutive channel

Let

$$\Gamma_\mu$$

denote an effective action obtained after coarse graining to a scale μ . Its leading first-order gravitational sector may be written schematically as

$$\Gamma_\mu^{\text{grav}} = \frac{1}{2} \int C_\mu(B)_{ab} \wedge R^{ab} + \Delta\Gamma_\mu, \quad (344)$$

where

$$B = e \wedge e \quad (345)$$

and $\Delta\Gamma_\mu$ contains operators outside the leading local algebraic constitutive channel.

If the leading algebraic term remains local and Lorentz equivariant and no additional background tensor is introduced into this channel, then

$$C_\mu = \alpha(\mu)I + \beta(\mu) \star. \quad (346)$$

For

$$\alpha(\mu)^2 + \beta(\mu)^2 \neq 0,$$

define

$$K(\mu) = \sqrt{\alpha(\mu)^2 + \beta(\mu)^2} > 0 \quad (347)$$

and

$$C_\mu = K(\mu)e^{\theta(\mu)\star}, \quad (348)$$

where

$$\alpha(\mu) = K(\mu) \cos \theta(\mu), \quad \beta(\mu) = K(\mu) \sin \theta(\mu). \quad (349)$$

The magnitude $K(\mu)$ is the scale-dependent constitutive stiffness, whereas $\theta(\mu)$ specifies the constitutive orientation inside the local Lorentz-equivariant bivector plane.

The pure Hodge branches are

$$\theta = \varepsilon_C \frac{\pi}{2} \pmod{2\pi}, \quad \varepsilon_C = \pm 1. \quad (350)$$

Definition 7.1 (Constitutive closure hierarchy). *Consider a coarse-grained gravitational response at scale μ .*

We distinguish:

1. Algebraic Lorentz-channel closure: *the leading local algebraic bivector constitutive operator is of the form*

$$C_\mu = \alpha(\mu)I + \beta(\mu)\star.$$

2. Effective constitutive-plane closure: *after the universal tensor and derivative structures of the physical low-energy source-to-response channel have been removed, the relevant response can still be parametrized by two scalar constitutive form factors multiplying I and $\star$.*
3. Einstein-ray closure: *the effective local constitutive response remains on one pure Hodge ray,*

$$C_\mu = \varepsilon_C K(\mu)\star,$$

throughout the range of scales under consideration.

The first statement follows from the local commutant classification only after projection onto the algebraic bivector channel. The second and third are dynamical properties.

In particular,

$$\text{End}_{SO^+(1,3)}(\Lambda^2 V) = \mathbb{R}I \oplus \mathbb{R}\star \quad (351)$$

does not exclude effective operators schematically of the form

$$R \wedge \star R, \quad T \wedge \star T, \quad R f(-D^2/\mu^2) R, \quad (352)$$

nor more general nonlocal kernels.

The constitutive plane is therefore a distinguished low-derivative projection of the effective response rather than the complete space of quantum corrections.

7.2 Radial and angular renormalization flow

Suppose the projected local coefficients obey

$$\mu \frac{d\alpha}{d\mu} = b_\alpha(\alpha, \beta, \dots), \quad (353)$$

and

$$\mu \frac{d\beta}{d\mu} = b_\beta(\alpha, \beta, \dots), \quad (354)$$

where the omitted arguments denote additional running effective couplings.

Differentiating

$$K^2 = \alpha^2 + \beta^2$$

gives

$$\mu \frac{dK}{d\mu} = \cos \theta b_\alpha + \sin \theta b_\beta. \quad (355)$$

Similarly,

$$\mu \frac{d\theta}{d\mu} = \frac{\cos \theta b_\beta - \sin \theta b_\alpha}{K}, \quad K \neq 0. \quad (356)$$

Thus the projected flow separates naturally into a radial stiffness flow and an angular constitutive flow.

Near a pure Hodge branch it is useful, whenever $\beta \neq 0$, to introduce

$$\chi := \frac{\alpha}{\beta}. \quad (357)$$

The pure Hodge ray is

$$\chi = 0.$$

Its flow is

$$\mu \frac{d\chi}{d\mu} = \frac{\beta b_\alpha - \alpha b_\beta}{\beta^2}. \quad (358)$$

Proposition 7.2 (Einstein-ray invariance criterion). *Within an effective constitutive-plane truncation, the pure Hodge ray*

$$\alpha = 0, \quad \beta \neq 0$$

is invariant under the renormalization flow if

$$b_\alpha(0, \beta, \dots) = 0 \quad (359)$$

along that ray.

This separates two logically different statements. Representation theory determines the available local invariant channels I and $\star$. It does not imply that the renormalization flow preserves

$$\alpha = 0.$$

Such preservation requires a dynamical mechanism or a symmetry that forbids generation of the identity channel.

7.3 Exact complex-comparison closure

Recall the doubled constitutive quarter-turn

$$J_\zeta(B, H) = (\zeta H, -\zeta^{-1} B), \quad J_\zeta^2 = -I. \quad (360)$$

Exact invariance of the constitutive graph

$$H = C(B)$$

under J_ζ is equivalent to

$$C^2 = -\zeta^{-2} I. \quad (361)$$

For

$$C = \alpha I + \beta \star,$$

one obtains

$$C^2 = (\alpha^2 - \beta^2) I + 2\alpha\beta \star. \quad (362)$$

Hence exact graph invariance requires

$$2\alpha\beta = 0 \quad (363)$$

and

$$\alpha^2 - \beta^2 = -\zeta^{-2}. \quad (364)$$

For real coefficients and a nonzero response,

$$\alpha = 0, \quad \beta = \varepsilon_C \zeta^{-1}, \quad \varepsilon_C = \pm 1. \quad (365)$$

Thus the exact graph-invariance statement remains unchanged.

Define the exact dimensional defect

$$D_\zeta(C) := C^2 + \zeta^{-2} I. \quad (366)$$

Then

$$D_\zeta(C) = 0 \quad (367)$$

is exactly equivalent to the graph condition Eq. (361).

However, the unnormalized condition

$$D_{\zeta(\mu)}(C_\mu) \longrightarrow 0 \quad (368)$$

is *not*, by itself, a valid criterion for convergence of the constitutive direction if the overall magnitude $K(\mu)$ is allowed to vanish.

Indeed, if $C_\mu \rightarrow 0$ and simultaneously $\zeta(\mu)^{-1} \rightarrow 0$, then Eq. (368) may hold even when the angle $\theta(\mu)$ remains away from the Hodge rays.

The infrared direction must therefore be tested by a normalized quantity.

7.4 Normalized complex-comparison closure

Whenever

$$K(\mu) > 0,$$

define the normalized constitutive operator

$$\widehat{C}_\mu := \frac{C_\mu}{K(\mu)} = \cos \theta(\mu) I + \sin \theta(\mu) \star. \quad (369)$$

It satisfies

$$\|\widehat{C}_\mu\|_{\text{constitutive}} = 1$$

in the coefficient norm associated with the basis $(I, \star)$.

Define the normalized directional defect

$$\widehat{D}(C_\mu) := \widehat{C}_\mu^2 + I. \quad (370)$$

Using $\star^2 = -I$,

$$\widehat{C}_\mu^2 = \cos 2\theta(\mu) I + \sin 2\theta(\mu) \star, \quad (371)$$

and therefore

$$\widehat{D}(C_\mu) = [1 + \cos 2\theta(\mu)] I + \sin 2\theta(\mu) \star. \quad (372)$$

Consequently,

$$\widehat{D}(C_\mu) = 0 \iff \theta(\mu) = \frac{\pi}{2} \pmod{\pi}. \quad (373)$$

Proposition 7.3 (Normalized pure-Hodge criterion). *Let*

$$C_\mu = K(\mu)e^{\theta(\mu)\star}, \quad K(\mu) > 0.$$

Then

$$\widehat{C}_\mu^2 + I \longrightarrow 0 \quad (374)$$

if and only if the distance of $\theta(\mu)$ from the union of the two pure Hodge rays tends to zero:

$$\text{dist}\left(\theta(\mu), \frac{\pi}{2} + \pi\mathbb{Z}\right) \longrightarrow 0. \quad (375)$$

The normalized condition determines the Hodge *direction* but not its branch. To select a fixed branch $\varepsilon_C = \pm 1$, one may additionally require

$$\frac{\beta(\mu)}{K(\mu)} \longrightarrow \varepsilon_C. \quad (376)$$

Equivalently,

$$\theta(\mu) \longrightarrow \varepsilon_C \frac{\pi}{2} \pmod{2\pi}. \quad (377)$$

Definition 7.4 (Complex-comparison infrared directional closure). *An effective gravitational response exhibits complex-comparison infrared directional closure if, after projection onto a nonzero physical local constitutive channel,*

$$\lim_{\mu \rightarrow 0} (\widehat{C}_\mu^2 + I) = 0. \quad (378)$$

If a definite branch is required, one also specifies Eq. (376).

This definition is insensitive to the overall magnitude of the constitutive response. A complete Einstein infrared target requires a separate stiffness condition:

$$K(\mu) \longrightarrow K_G > 0. \quad (379)$$

Thus the appropriate infrared target is

$$\theta_{\text{op}}(\mu) \longrightarrow \varepsilon_C \frac{\pi}{2}, \quad K_{\text{op}}(\mu) \longrightarrow K_G > 0. \quad (380)$$

The two conditions should not be conflated:

$$\text{directional closure} \quad \neq \quad \text{stiffness determination}. \quad (381)$$

Moreover, because bare action parameters can be affected by field redefinitions, topological rearrangements, and renormalization conventions, the most robust infrared statement should preferentially be formulated using the operational response decomposition rather than only a bare Holst or Immirzi parameter.

7.5 Coupled comparison sectors

The effective response problem becomes richer when the gravitational channel couples to another physical comparison sector.

Let X denote a physical gravitational mode after gauge reduction and let Y denote a physical mode in an internal or relative-comparison sector.

At quadratic order, a finite-dimensional static or Euclidean truncation may be written as

$$\Gamma_{\text{stat}}^{(2)} = \frac{1}{2} \begin{pmatrix} X^\dagger & Y^\dagger \end{pmatrix} \begin{pmatrix} K_G & M \\ M^\dagger & K_I \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}. \quad (382)$$

Here K_G and K_I are the gravitational and internal stiffness matrices and M is the quadratic mixing block.

In field theory these symbols may originate from differential kernels. The algebra below is exact for finite-dimensional mode truncations and, with the assumptions stated below, for bounded operators. Unbounded differential operators require a corresponding formulation in terms of closed quadratic forms and their domains.

Suppose first that K_I is invertible. Stationarity with respect to Y at fixed X gives

$$Y_{\text{stat}} = -K_I^{-1} M^\dagger X. \quad (383)$$

Substitution gives

$$\Gamma_{\text{stat,eff}}^{(2)}[X] = \frac{1}{2} X^\dagger K_G^{\text{eff}} X, \quad (384)$$

where

$$K_G^{\text{eff}} = K_G - M K_I^{-1} M^\dagger. \quad (385)$$

The second term is the Schur-complement correction.

Theorem 7.5 (Static comparison-mixing softening theorem). *Let*

$$\mathcal{H}_G, \quad \mathcal{H}_I$$

be finite-dimensional Hilbert spaces, or Hilbert spaces on which K_G, K_I, M define bounded operators. Assume

$$K_G = K_G^\dagger, \quad K_I = K_I^\dagger, \\ K_I > 0,$$

and

$$\mathbb{K} := \begin{pmatrix} K_G & M \\ M^\dagger & K_I \end{pmatrix} > 0. \quad (386)$$

Then

$$K_G^{\text{eff}} = K_G - MK_I^{-1}M^\dagger > 0 \quad (387)$$

and

$$K_G^{\text{eff}} \leq K_G \quad (388)$$

in the ordering of Hermitian quadratic forms.

Proof. Since $K_I > 0$, the positive square root $K_I^{-1/2}$ exists. For every X ,

$$\langle X, MK_I^{-1}M^\dagger X \rangle = \|K_I^{-1/2}M^\dagger X\|^2 \geq 0. \quad (389)$$

Hence

$$MK_I^{-1}M^\dagger \geq 0.$$

Positivity of the block matrix together with $K_I > 0$ is equivalent to positivity of its Schur complement,

$$K_G - MK_I^{-1}M^\dagger > 0.$$

The second inequality follows immediately by subtracting a positive semidefinite operator from K_G . $\square$

Thus, in this positive static setting, allowing the additional comparison sector to relax can only reduce the effective stiffness of the gravitational mode:

$$K_G^{\text{eff}} \leq K_G. \quad (390)$$

Equivalently, its static compliance is enhanced on the corresponding positive physical channel.

For genuinely unbounded differential operators, the preceding theorem should not be used merely as a formal matrix identity. One must specify a common dense form domain and require, for example, that K_I define a strictly positive self-adjoint operator and that the mixed term be sufficiently form bounded so that

$$MK_I^{-1}M^\dagger$$

or its quadratic-form analogue is well defined. The Schur complement is then most naturally formulated at the level of closed quadratic forms.

7.6 Single-mode stability bound

For one normalized mode in each sector,

$$\Gamma_{\text{stat}}^{(2)} = \frac{1}{2} \begin{pmatrix} X^* & Y^* \end{pmatrix} \begin{pmatrix} K_G & m \\ m^* & K_I \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}, \quad (391)$$

one obtains

$$K_G^{\text{eff}} = K_G - \frac{|m|^2}{K_I}. \quad (392)$$

If

$$K_G > 0, \quad K_I > 0,$$

positivity of the full two-mode Hessian is equivalent to

$$|m|^2 < K_G K_I. \quad (393)$$

Corollary 7.6 (Comparison-mixing stability bound). *In a strictly stable two-mode static response sector,*

$$|m| < \sqrt{K_G K_I}. \quad (394)$$

Consequently,

$$K_G^{\text{eff}} < K_G \quad (395)$$

for nonzero mixing, while

$$\frac{1}{K_G^{\text{eff}}} > \frac{1}{K_G} \quad (396)$$

as long as the effective stiffness remains positive.

The physical interpretation is that part of the deformation generated by a source applied to the gravitational mode may relax through the coupled comparison mode.

7.7 Approach to a soft comparison mode

The stability inequality becomes especially restrictive when the additional sector becomes soft.

Suppose

$$K_I \longrightarrow 0^+ \quad (397)$$

while

$$K_G \longrightarrow K_0 > 0. \quad (398)$$

Then

$$|m|^2 < K_G K_I$$

requires

$$|m|^2 = O(K_I) \quad (399)$$

or smaller, equivalently

$$|m| = O(\sqrt{K_I}) \quad (400)$$

or smaller.

Proposition 7.7 (Soft-sector consistency condition). *If an internal comparison stiffness tends continuously to zero while the combined static two-mode system remains strictly positive and the gravitational stiffness remains finite and nonzero, then the direct quadratic mixing must satisfy*

$$|m|^2 < K_G K_I \quad (401)$$

and therefore must vanish at least at the scale

$$|m| = O(\sqrt{K_I})$$

or faster.

Otherwise the coupled Hessian reaches a zero mode or an instability before the internal sector becomes independently soft.

This provides a useful consistency test for proposed emergent comparison sectors. If an internal mode becomes massless or highly compliant in the infrared while gravity remains in a regular positive static phase, its direct mixing with the gravitational static mode must be correspondingly suppressed.

7.8 Real-time response and the limits of static softening

The positive softening theorem applies only to a static, Euclidean, or reduced positive energy Hessian.

A general retarded inverse response

$$\Gamma_R(\omega, \mathbf{k})$$

is frequency dependent and generally complex. It does not possess a global ordering analogous to the positive static quadratic-form ordering.

For a coupled retarded response matrix

$$\Gamma_R = \begin{pmatrix} \Gamma_{GG,R} & \Gamma_{GI,R} \\ \Gamma_{IG,R} & \Gamma_{II,R} \end{pmatrix}, \quad (402)$$

formal elimination of the internal sector gives

$$\Gamma_{GG,R}^{\text{eff}} = \Gamma_{GG,R} - \Gamma_{GI,R} \Gamma_{II,R}^{-1} \Gamma_{IG,R}, \quad (403)$$

where the inverse and products must be understood on the appropriate causal response domain.

The correction may alter

dispersion, damping, pole positions, residues.

It should not, in general, be described as a positive softening.

Thus

$$\text{static Schur softening} \not\Rightarrow \text{monotonic softening of the full retarded kernel.} \quad (404)$$

7.9 General response matrix

For several coupled response channels, collect the physical modes and sources into

$$\mathcal{A} = \begin{pmatrix} \mathcal{A}_G \\ \mathcal{A}_I \end{pmatrix}, \quad J = \begin{pmatrix} J_G \\ J_I \end{pmatrix}. \quad (405)$$

The linearized response equation is

$$\Gamma^{(2)} \delta \mathcal{A} = \delta J, \quad (406)$$

with

$$\Gamma^{(2)} = \begin{pmatrix} \Gamma_{GG} & \Gamma_{GI} \\ \Gamma_{IG} & \Gamma_{II} \end{pmatrix}. \quad (407)$$

Its inverse

$$K = (\Gamma^{(2)})^{-1} = \begin{pmatrix} K_{GG} & K_{GI} \\ K_{IG} & K_{II} \end{pmatrix} \quad (408)$$

contains both diagonal and cross susceptibilities.

The off-diagonal kernel

$$K_{GI}$$

measures the response of a gravitational observable to an internal source, whereas

$$K_{IG}$$

measures the response of an internal comparison observable to a gravitational source.

These cross kernels provide an operational characterization of sector mixing.

7.10 Symmetry obstruction to direct mixing

The existence of a quadratic mixing block is constrained by symmetry.

Suppose the unbroken local symmetry is a direct product

$$\text{Spin}^+(1, 3) \times G_{\text{int}}. \quad (409)$$

Let R_G and R_I denote the representations carried by the physical gravitational and internal comparison modes.

A quadratic mixing operator

$$M : R_I \longrightarrow R_G$$

must be an invariant intertwiner.

Proposition 7.8 (Symmetry obstruction to direct comparison mixing). *A direct quadratic mixing term between modes transforming in representations R_G and R_I can exist only if*

$$\text{Hom}_{\text{Spin}^+(1,3) \times G_{\text{int}}}(R_I, R_G) \neq 0. \quad (410)$$

If the gravitational mode is a G_{int} -singlet and R_I contains no G_{int} -singlet subrepresentation, then

$$M = 0 \quad (411)$$

as long as the product symmetry remains unbroken.

Thus generic direct bilinear mixing between a purely geometric singlet mode and an adjoint mode of a simple unbroken internal group is forbidden.

Nonzero mixing may nevertheless occur when additional representation structure supplies an allowed intertwiner, for example through an appropriate singlet channel, an Abelian sector, a mediator, a symmetry-breaking order parameter, or a background that reduces the unbroken product symmetry.

The precise mechanism is model dependent.

The important point is that the existence of two connection-like variables does not by itself supply such an intertwiner.

For example, a total covariant derivative may take the tensor-product form

$$\nabla = \nabla_{\text{geom}} \otimes I + I \otimes \nabla_{\text{int}}, \quad (412)$$

without implying a direct invariant quadratic mixing term between the two connection sectors.

7.11 Noise transfer through an allowed mixed sector

When mixing is symmetry allowed, fluctuations in the additional comparison sector can feed into the gravitational observable.

Schematically, suppose the linearized gravitational response contains

$$X = X_0 - K_G^{-1} M Y + \dots \quad (413)$$

If Y has noise kernel N_I , then the induced contribution to the gravitational noise takes the schematic form

$$N_G^{\text{ind}} \sim K_G^{-1} M N_I M^\dagger (K_G^{-1})^\dagger, \quad (414)$$

in the static algebraic approximation.

In a causal open-system treatment, the corresponding expression contains retarded and advanced response kernels rather than simple algebraic inverse stiffnesses:

$$N_G^{\text{ind}} \sim K_{GI}^R N_I K_{IG}^A, \quad (415)$$

schematically.

Thus a nongeometric comparison sector can contribute to geometric fluctuations without becoming identical to the Lorentz geometric sector:

$$\text{internal contribution to geometric noise} \not\Rightarrow \text{identity of the internal and Lorentz sectors.} \quad (416)$$

7.12 Scale dependence of sector mixing

Let the full inverse response matrix depend on the coarse-graining scale:

$$\Gamma_\mu^{(2)} = \begin{pmatrix} \Gamma_{GG,\mu} & \Gamma_{GI,\mu} \\ \Gamma_{IG,\mu} & \Gamma_{II,\mu} \end{pmatrix}. \quad (417)$$

Even if

$$\Gamma_{GI,\mu_0} = 0$$

at some reference scale μ_0 , radiative corrections may generate a cross block unless an exact unbroken symmetry forbids the required intertwiner.

Likewise, even if

$$C_{\mu_0} = K(\mu_0)\star \quad (418)$$

at one scale, the identity channel may be regenerated unless the pure Hodge ray is protected dynamically or by symmetry.

The strongest useful infrared closure condition therefore combines several requirements:

$$\theta_{\text{op}}(\mu) \longrightarrow \varepsilon_C \frac{\pi}{2}, \quad (419)$$

$$K_{\text{op}}(\mu) \longrightarrow K_G > 0, \quad (420)$$

$$\Gamma_{GI,\mu} \longrightarrow 0 \quad \text{or a symmetry-controlled finite structure,} \quad (421)$$

and

$$\Delta\Gamma_\mu \longrightarrow \text{infrared-irrelevant corrections} \quad (422)$$

relative to the leading gravitational response.

Definition 7.9 (Infrared Einstein comparison closure). *An effective MS gravitational sector exhibits infrared Einstein comparison closure if, on the physical long-wavelength response space,*

1. *the operational constitutive direction approaches one pure Hodge branch;*
2. *the gravitational stiffness approaches a finite positive value;*
3. *additional local, higher-derivative, and nonlocal response channels become irrelevant or parametrically suppressed relative to the leading Einstein channel; and*
4. *mixing with nongeometric comparison sectors either vanishes or reduces to symmetry-compatible effects that can be absorbed into finite renormalizations or explicitly retained controlled response channels.*

This definition isolates the dynamical statement required to pass from the broad MS comparison structure to an ordinary low-energy Einstein-like response.

The local rigidity theorem constrains the possible algebraic endpoint of such a flow. It does not prove that the renormalization flow reaches that endpoint.

The logical structure is therefore

$$\begin{aligned} \text{Lorentz representation theory} &\longrightarrow \mathbb{R}I \oplus \mathbb{R}\star, \\ \text{comparison--constitutive compatibility} &\longrightarrow \text{pure-Hodge candidate direction,} \\ \text{renormalization flow} &\longrightarrow K(\mu), \theta(\mu), \Gamma_{GI,\mu}, \dots, \\ \text{infrared dynamics} &\longrightarrow \text{possible Einstein comparison closure.} \end{aligned} \quad (423)$$

The first two statements are structural and conditional on the assumptions already made. The last two are dynamical.

7.13 Einstein-direction rigidity versus Newton-stiffness rigidity

The separation between constitutive direction and stiffness remains essential in the infrared.

Even if

$$\theta_{\text{op}} \longrightarrow \varepsilon_C \frac{\pi}{2}, \quad (424)$$

the low-energy magnitude

$$K_G$$

remains a renormalized response coefficient.

Mixing with massive comparison sectors can shift it through Schur-complement corrections. Microscopic quantum fluctuations can modify it through local counterterms and nonlocal form factors.

Hence

$$\text{Einstein-direction rigidity} \not\Rightarrow \text{Newton-stiffness rigidity.} \quad (425)$$

A microscopic MS model would have to explain both independently.

7.14 Infrared failure modes

The preceding analysis identifies several qualitatively different ways in which Einstein comparison closure may fail.

If

$$\theta_{\text{op}}(\mu)$$

does not approach a pure Hodge branch, the infrared response retains two independent Lorentz-equivariant algebraic constitutive channels.

If higher-derivative or nonlocal corrections remain comparable to the leading low-energy channel, an approximately local two-derivative gravitational response does not emerge.

If an additional comparison sector becomes soft while its mixing with the gravitational mode violates

$$|m|^2 < K_G K_I,$$

the assumed positive static vacuum develops a zero mode or instability before the internal sector becomes independently soft.

If the additional sector remains massive and sufficiently stiff, integrating it out can instead produce finite local or weakly nonlocal renormalizations of the gravitational response.

These alternatives are physically distinct and should not be collapsed into one undifferentiated statement of “emergent gravity.”

7.15 Operational response-matrix criterion

A useful low-energy diagnostic is the complete matrix of retarded susceptibilities,

$$K_R(\omega, \mathbf{k}) = \begin{pmatrix} K_{GG,R} & K_{GI,R} \\ K_{IG,R} & K_{II,R} \end{pmatrix}, \quad (426)$$

together with the corresponding symmetrized noise matrix

$$N(\omega, \mathbf{k}) = \begin{pmatrix} N_{GG} & N_{GI} \\ N_{IG} & N_{II} \end{pmatrix}, \quad (427)$$

when spatial translation invariance is present. In an inhomogeneous background the corresponding kernels should retain their position arguments.

These matrices distinguish

$$\text{gravitational response,} \quad \text{internal comparison response,} \quad \text{cross-sector transduction.} \quad (428)$$

In an equilibrium KMS state, the corresponding matrix-valued fluctuation–dissipation relation constrains the response and noise matrices simultaneously, subject to the operator-index and Fourier conventions discussed in Section 6.

An operational test of Einstein-channel closure can therefore be formulated without relying exclusively on a chosen bare action.

One may ask whether the measured long-wavelength response matrix contains a gravitational eigenchannel for which

$$\theta_{\text{op}} \longrightarrow \varepsilon_C \frac{\pi}{2}, \quad (429)$$

$$\Gamma_{GG,R}^{(2)} \longrightarrow K_G D_G \quad (430)$$

after gauge reduction and normalization, while

$$K_{GI,R}, K_{IG,R} \longrightarrow \text{symmetry-suppressed or controlled forms}, \quad (431)$$

and higher-derivative corrections vanish relative to the leading long-wavelength response.

This formulation is less dependent on a particular bare parametrization and follows the operational logic of the MS reconstruction more closely.

7.16 Summary

The effective response structure developed in this section may be summarized as

$$C_\mu = K(\mu) e^{\theta(\mu)\star}, \quad (432)$$

$$\theta(\mu) \longrightarrow \text{constitutive orientation flow}, \quad (433)$$

$$K(\mu) \longrightarrow \text{gravitational stiffness flow}, \quad (434)$$

$$M \longrightarrow \text{cross-comparison mixing}, \quad (435)$$

and, in a positive static finite-mode channel,

$$K_G^{\text{eff}} = K_G - M K_I^{-1} M^\dagger. \quad (436)$$

The corrected infrared directional criterion is

$$\widehat{C}_\mu = \frac{C_\mu}{K(\mu)}, \quad \widehat{C}_\mu^2 + I \longrightarrow 0, \quad (437)$$

together, for a complete Einstein infrared response, with

$$K(\mu) \longrightarrow K_G > 0. \quad (438)$$

Thus

$$\text{local rigidity} + \text{dynamical directional closure} + \text{finite positive stiffness} + \text{controlled sector mixing} \quad (439)$$

defines the candidate low-energy Einstein comparison regime.

The representation-theoretic results constrain the available local constitutive directions. They do not determine the renormalization trajectory, the value of Newton’s stiffness, or the fate of additional comparison sectors. Those remain dynamical questions for a microscopic MS theory.

8 Discussion, Limitations, and Outlook

The purpose of the present paper has been to investigate whether several structures that appear separately in the motion–staticity (MS) reconstruction program can be related without identifying them prematurely.

The central structures considered here are

the complex structure of quantum comparison,
the Hodge structure of Lorentz bivectors,
the constitutive response of first-order gravity,

and

the operator-valued geometric response to quantum sources.

The principal result is not that these objects are identical. They live on different representation spaces and occur at different logical levels. Rather, under explicitly stated compatibility assumptions, they can be connected through a sequence of representation-theoretic, topological, constitutive, and response-theoretic maps.

The distinction between exact mathematical results and additional physical assumptions is essential throughout.

8.1 Spinorial complex structure and Hodge duality

The first important distinction concerns the role of the scalar imaginary unit.

A complex quantum carrier possesses scalar multiplication by i , but this fact alone does not imply a Hodge operator on spacetime geometry. Several different complex structures appear in the construction:

$$\psi \mapsto i\psi \tag{440}$$

on a complex spinor or quantum carrier,

$$X \mapsto iX \tag{441}$$

on $\mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}}$, and

$$B \mapsto \star B \tag{442}$$

on the real Lorentz bivector representation.

These are not literally the same operator.

The representation-theoretic bridge becomes available when the Lorentz spin-cover algebra is related to the gravitational Lorentz-generator sector through

$$d\pi : \mathfrak{sl}(2, \mathbb{C})_{\mathbb{R}} \xrightarrow{\cong} \mathfrak{so}(1, 3) \simeq \Lambda^2 V. \tag{443}$$

Multiplication by i on the complex Lie algebra defines the real-linear complex structure

$$J_{\text{spin}} X = iX, \quad J_{\text{spin}}^2 = -I. \tag{444}$$

Transporting it gives

$$J_B = d\pi J_{\text{spin}} d\pi^{-1}. \tag{445}$$

In four spacetime dimensions,

$$\text{End}_{SO^+(1,3)}(\Lambda^2 V) = \mathbb{R}I \oplus \mathbb{R}\star, \quad (446)$$

and

$$\star^2 = -I. \quad (447)$$

Consequently, the only Lorentz-equivariant real complex structures on the bivector carrier are

$$J_B = \pm \star. \quad (448)$$

The precise statement is therefore

$$X \mapsto iX \xrightarrow{d\pi} J_B = \pm \star, \quad (449)$$

not the literal identity

$$i = \star.$$

In chiral language,

$$\Lambda^2 V_{\mathbb{C}} = \Lambda_+^2 \oplus \Lambda_-^2, \quad (450)$$

with

$$\star X_{\pm} = \pm i X_{\pm}, \quad (451)$$

and, up to the conventional assignment of chirality,

$$\Lambda_+^2 \simeq \text{Sym}^2 S_L, \quad \Lambda_-^2 \simeq \text{Sym}^2 S_R. \quad (452)$$

The mathematical relation is exact once the Lorentz spin-cover identification has been specified.

The physical identification remains conditional. A projective quantum comparison structure does not by itself prove that the dynamically realized quantum carrier and the reconstructed spacetime geometry use the same local $\text{Spin}^+(1, 3)$ representation.

Accordingly,

$$\text{nontrivial projective rotational class} \not\Rightarrow J_B = \pm \star \quad (453)$$

without the additional common-spin-cover compatibility condition.

The present paper determines what follows once such a common realization is assumed. A microscopic comparison theory must explain why that common lift is dynamically selected.

8.2 Four-dimensional bulk rigidity

The second major result concerns spacetime dimension.

The generalized MS state may be written without fixing the number of spatial components in advance:

$$D \in \mathbb{R}^m, \quad |D|^2 + S^2 = 1, \quad (454)$$

with

$$d = m + 1. \quad (455)$$

The gravitational generator carrier is

$$W_d = \Lambda^2 \mathbb{R}^{1,d-1} \simeq \mathfrak{so}(1, d-1). \quad (456)$$

Within the one-time Lorentz family considered in this paper, requiring an observer-equivariant real complex structure

$$J_G \in \text{End}_{SO^+(1,d-1)}(W_d), \quad J_G^2 = -I, \quad (457)$$

selects

$$d = 4, \quad J_G = \pm \star. \quad (458)$$

This result does not arise merely from observing that the Hodge operator maps two-forms to two-forms in four dimensions. The argument first classifies the complete equivariant endomorphism algebra of the Lorentz bivector carrier and only afterwards determines whether that real algebra contains a complex structure.

The conclusion must nevertheless retain its representation-theoretic scope:

$$\begin{aligned} &\text{Lorentz-equivariant complex structure on the} \\ &\text{adjoint/bivector gravitational carrier} \implies d = 4. \end{aligned} \quad (459)$$

It is not a theorem that every conceivable gravitational carrier or every possible theory of quantum geometry must live in four dimensions.

8.3 Independent null-boundary rigidity

A second dimensional mechanism arises from the projective null boundary.

For m spatial dimensions, the null-direction sphere is

$$S^{m-1} = S^{d-2}. \quad (460)$$

Suppose that the fundamental finite-helicity boundary degree of freedom is represented by a complex line bundle

$$L \longrightarrow S^{d-2} \quad (461)$$

with

$$c_1(L) \neq 0. \quad (462)$$

Then

$$H^2(S^{d-2}; \mathbb{Z}) \neq 0.$$

For a sphere this occurs only for

$$d - 2 = 2,$$

and therefore

$$d = 4. \quad (463)$$

This boundary mechanism is mathematically independent of the bulk commutant argument. The two statements may be summarized as

$$\begin{aligned} &\text{bulk bivector-complex requirement} \implies d = 4, \\ &\text{boundary nontrivial helicity-line requirement} \implies d = 4. \end{aligned} \quad (464)$$

Their agreement provides a conditional bulk–boundary double rigidity.

The assumptions of the boundary theorem are essential. Higher-dimensional massless theories can possess non-Abelian little-group representations, higher-rank polarization bundles, and other topological structures. The argument does not exclude those possibilities. It establishes rigidity only for the rank-one complex line-bundle mechanism with nonzero first Chern class.

In four dimensions, the standard half-helicity line satisfies

$$|c_1(L_{1/2})| = 1. \quad (465)$$

Since

$$S^2 \simeq \mathbb{CP}^1,$$

its bundle type is, depending on orientation and helicity conventions,

$$L_{1/2} \simeq \mathcal{O}(-1) \quad \text{or} \quad \mathcal{O}(1). \quad (466)$$

This produces the familiar network

$$S_L \simeq \mathbb{C}^2, \quad \mathbb{P}(S_L) \simeq \mathbb{CP}^1 \simeq S^2, \quad (467)$$

together with

$$\text{Herm}_2(\mathbb{C}) \simeq \mathbb{R}^{1,3}. \quad (468)$$

These are mathematical correspondences, not ontological identifications. A spinor ray, a celestial null direction, and a helicity line are different objects even when their homogeneous spaces or bundle classes are naturally isomorphic.

The MS-specific problem is therefore not to rederive these standard isomorphisms, but to determine whether one dynamically selected comparison carrier realizes them coherently within a single microscopic construction.

8.4 Constitutive complex structure

The third major result concerns the gravitational constitutive map.

For a local real-linear Lorentz-equivariant response on four-dimensional bivectors,

$$C = \alpha I + \beta \star. \quad (469)$$

The invariant constitutive algebra is

$$\mathbb{R}I \oplus \mathbb{R}\star \simeq \mathbb{C}. \quad (470)$$

For

$$(\alpha, \beta) \neq (0, 0),$$

one may write

$$C = Ke^{\theta\star}, \quad K > 0. \quad (471)$$

The two parameters have distinct meanings:

K = constitutive magnitude or stiffness,

while

θ = constitutive orientation.

The doubled constitutive response space

$$\mathcal{R}_G = W \oplus W$$

admits the real complex structure

$$J_\zeta(B, H) = (\zeta H, -\zeta^{-1} B), \quad J_\zeta^2 = -I. \quad (472)$$

If the physical constitutive graph

$$\Gamma_C = \{(B, CB)\}$$

is invariant under this transformation, then

$$C^2 = -\zeta^{-2} I. \quad (473)$$

Inside $\mathbb{R}I \oplus \mathbb{R}\star$, the real nonzero solutions are

$$C = \pm \zeta^{-1} \star. \quad (474)$$

Thus graph invariance selects the pure Hodge constitutive direction.

The logical status of this result must remain explicit. The doubled pair (B, H) is a constitutive input–output space. It is not automatically a canonical gravitational phase space, and J_ζ is not derived here as the canonical complex structure of a fully quantized gravitational system.

Accordingly, the implication is

$$\begin{aligned} &\text{comparison complex structure represented as } J_\zeta \\ &+ \text{constitutive graph invariance} \\ &\implies C \propto \star. \end{aligned} \quad (475)$$

A microscopic theory must explain why the primitive comparison complex structure acts on the gravitational constitutive variables in precisely this form.

8.5 Source-induced operator geometry

The fourth major result concerns the quantum status of matter-induced geometry.

For an invertible algebraic torsional response,

$$C \mathcal{T}_e(T) = \tau, \quad (476)$$

one may write

$$T = L_e \tau, \quad L_e = \mathcal{T}_e^{-1} C^{-1}. \quad (477)$$

Promoting the matter source to an operator gives the source-induced geometric observable

$$\widehat{T} = L_e \widehat{\tau}. \quad (478)$$

For any fixed multilinear operator-ordering prescription O , the connected correlators satisfy

$$\begin{aligned} &\langle O \widehat{T}(x_1) \cdots \widehat{T}(x_n) \rangle_c \\ &= \left(\bigotimes_{r=1}^n L_{x_r} \right) \langle O \widehat{\tau}(x_1) \cdots \widehat{\tau}(x_n) \rangle_c. \end{aligned} \quad (479)$$

Likewise,

$$[\widehat{T}(x), \widehat{T}(y)] = L_x[\widehat{\tau}(x), \widehat{\tau}(y)]L_y^T \quad (480)$$

in the local real-response notation.

Consequently,

$$\langle \widehat{T} \rangle = 0 \quad (481)$$

does not imply

$$\langle \widehat{T}(x)\widehat{T}(y) \rangle_c = 0. \quad (482)$$

A geometry whose mean torsion vanishes may therefore possess nontrivial source-induced quantum torsional correlations.

This is operator-valued quantum geometric response, but it is not by itself an independently quantized gravitational field:

$$\text{operator-valued geometry} \not\Rightarrow \text{intrinsic quantum gravity}. \quad (483)$$

The generating-functional formulation requires an additional operator-ordering qualification.

For the ordinary characteristic functional

$$Z_\tau^W[J] = \left\langle \exp \left[\frac{i}{\hbar} \widehat{\tau}(J) \right] \right\rangle, \quad (484)$$

functional derivatives generate the Weyl/symmetrically ordered hierarchy. In the purely linear source-induced sector,

$$W_\tau^W[K] = W_\tau^W[L_e^\dagger K]. \quad (485)$$

Time-ordered correlators instead require

$$Z_\tau^T[J] = \left\langle \mathcal{T} \exp \left[\frac{i}{\hbar} \int J \cdot \widehat{\tau} \right] \right\rangle, \quad (486)$$

and general real-time Wightman orderings require an appropriate multi-source or Schwinger–Keldysh construction.

Thus there is no single ordinary generating functional that simultaneously generates every possible fixed ordering of noncommuting local operators.

The general diagnostic nevertheless remains: if all geometric quantum data in an effective channel are exhausted by the appropriately ordered pullback of the matter-source generating functional through L_e , then the geometric quantumness of that channel is source induced.

Intrinsic quantum geometry requires additional information that cannot be reconstructed from the source algebra and the deterministic response kernel alone.

Schematically,

$$C_{GG}^{\text{full}} = C_{GG}^{\text{source}} + C_{GG}^{\text{intrinsic}} + C_{GG}^{\text{mixed}}. \quad (487)$$

The precise decomposition is model dependent and must ultimately be defined at the level of a microscopic operator algebra, Hilbert-space factorization, or effective open-system description.

8.6 Matter entanglement and algebraic gravitational response

An algebraic torsion mode can be eliminated from the effective action, producing

$$S_{\text{eff}} \supset -\frac{1}{2} \int \tau K_T^{-1} \tau. \quad (488)$$

For suitable low-energy spin degrees of freedom, the interaction can contain a term schematically of the form

$$H_{\text{eff}} \sim J_T \mathbf{S}_A \cdot \mathbf{S}_B. \quad (489)$$

Such an interaction can entangle suitable initially separable spin states.

This does not imply that torsion propagates as an independent quantum field. A nondynamical or algebraically eliminated geometric response can mediate a local effective interaction without possessing an independent asymptotic quantum.

Therefore,

$$\text{geometry-mediated entangling interaction} \not\Rightarrow \text{independent propagating quantum-gravitational mediator.} \quad (490)$$

Additional assumptions concerning locality, independent mediator degrees of freedom, causal propagation, and operational measurement are required before stronger conclusions can be drawn.

8.7 Gravitational stiffness and Newton coupling

The fifth major result concerns the interpretation of gravitational coupling.

For the pure Hodge constitutive branch,

$$C = \varepsilon_C K_G \star. \quad (491)$$

Its inverse enters the algebraic source response as

$$T \propto K_G^{-1} \tau. \quad (492)$$

More generally, if a reduced infrared inverse response kernel factorizes as

$$\Gamma_R^{(2)} = K_G D_R, \quad (493)$$

then

$$K_R = K_G^{-1} D_R^{-1}. \quad (494)$$

Thus K_G is naturally interpreted as a gravitational stiffness, while K_G^{-1} is the corresponding compliance coefficient.

With conventional Einstein normalization,

$$K_G \propto \frac{c^3}{G}. \quad (495)$$

Newton's coupling therefore measures, up to normalization, the low-energy compliance of the corresponding gravitational response channel.

This is an operational interpretation, not a numerical derivation of Newton's constant.

The causal quantum response is described by

$$K_{AB}^R(x, y) = -\frac{i}{\hbar} \theta(x^0 - y^0) \langle [\widehat{O}_A(x), \widehat{O}_B(y)] \rangle, \quad (496)$$

with the overall sign depending on the source convention, while the symmetrized quantum noise kernel is

$$N_{AB}(x, y) = \frac{1}{2} \langle \{ \Delta \widehat{O}_A(x), \Delta \widehat{O}_B(y) \} \rangle. \quad (497)$$

In a KMS equilibrium state these quantities satisfy the fluctuation–dissipation relation.

A frequency-space representation requires stationarity. A simultaneous diagonal representation in $(\omega, \mathbf{k})$ additionally requires spatial translation invariance. On an inhomogeneous background one must instead retain kernels such as

$$K_R(\omega; \mathbf{x}, \mathbf{y})$$

or use an appropriate Wigner representation.

Causality and spectral information constrain the response but do not determine every real local coefficient. Subtracted dispersion relations contain subtraction constants and analytic local data that are not fixed by the absorptive spectrum.

Consequently,

$$\text{spectral response constraints} \not\Rightarrow \text{numerical prediction of } G. \quad (498)$$

A microscopic theory must still determine the finite renormalized gravitational stiffness.

8.8 Planck area as a response-conjugacy scale

The same gravitational stiffness has a second quantum interpretation.

In an appropriate reduced first-order gravitational channel, the presymplectic form has the schematic normalization

$$\Omega_{\text{grav}} \sim K_G \int_{\Sigma} \delta(\star B) \wedge_F \delta \omega. \quad (499)$$

After the constraints and gauge redundancies have been properly reduced, the corresponding canonical bracket scales as

$$\{\omega, \star B\}_{\text{red}} \sim K_G^{-1}. \quad (500)$$

Upon quantization,

$$[\widehat{\omega}, \widehat{\star B}]_{\text{red}} \sim i\hbar K_G^{-1}. \quad (501)$$

Hence

$$\hbar K_G^{-1} \quad (502)$$

sets the characteristic quantum conjugacy scale of this reduced gravitational response.

Since

$$K_G^{-1} \propto \frac{G}{c^3},$$

one obtains

$$\hbar K_G^{-1} \propto \frac{\hbar G}{c^3} = \ell_P^2. \quad (503)$$

The resulting interpretation is

$$\ell_p^2 \sim \hbar \times \text{gravitational compliance.} \quad (504)$$

This does not constitute a derivation of a minimum area. It identifies the Planck area as the natural dimensional scale obtained by combining the quantum action unit with gravitational compliance.

Whether a particular quantum theory possesses a discrete physical area spectrum, or a nonzero minimum eigenvalue, is a separate spectral question.

8.9 Effective constitutive flow

The sixth major result concerns coarse graining.

Within a projected local constitutive channel,

$$C_\mu = K(\mu)e^{\theta(\mu)\star}. \quad (505)$$

This separates two distinct flows:

$$K(\mu) : \text{stiffness flow}, \quad \theta(\mu) : \text{constitutive-orientation flow.} \quad (506)$$

The representation-theoretic classification of the local bivector commutant does not prove that the quantum effective response remains inside this two-dimensional truncation. Higher derivatives, nonlocal form factors, torsional operators, additional fields, and cross-sector mixing can be generated.

Thus

$$\text{four-dimensional local constitutive rigidity} \not\Rightarrow \text{renormalization-group closure.} \quad (507)$$

There is also an important distinction between an exact graph-invariance condition and an infrared directional limit.

For a nonzero constitutive operator define

$$\widehat{C}_\mu = \frac{C_\mu}{K(\mu)}. \quad (508)$$

A genuine infrared pure-Hodge directional condition is

$$\widehat{C}_\mu^2 + I \longrightarrow 0. \quad (509)$$

Equivalently,

$$\text{dist}\left(\theta(\mu), \frac{\pi}{2} + \pi\mathbb{Z}\right) \longrightarrow 0. \quad (510)$$

An unnormalized condition such as

$$C_\mu^2 + \zeta(\mu)^{-2}I \rightarrow 0$$

does not by itself determine the constitutive direction when $C_\mu \rightarrow 0$.

A complete infrared Einstein target therefore requires both

$$\theta_{\text{op}}(\mu) \longrightarrow \varepsilon_C \frac{\pi}{2} \quad (511)$$

and

$$K_{\text{op}}(\mu) \longrightarrow K_G > 0, \quad (512)$$

together with suppression or controlled renormalization of additional response channels.

The word “closure” is therefore essential. Representation theory constrains the possible local algebraic endpoint; it does not establish the renormalization trajectory.

8.10 Coupled comparison sectors

The coupled-response analysis gives an additional constraint.

For a reduced finite-mode static or Euclidean response, suppose

$$\mathbb{K} = \begin{pmatrix} K_G & M \\ M^\dagger & K_I \end{pmatrix}. \quad (513)$$

When

$$\mathbb{K} > 0, \quad K_I > 0,$$

in a finite-dimensional or appropriately bounded-operator setting, eliminating the internal mode gives the Schur-complement stiffness

$$K_G^{\text{eff}} = K_G - MK_I^{-1}M^\dagger. \quad (514)$$

Moreover,

$$0 < K_G^{\text{eff}} \leq K_G. \quad (515)$$

A stable additional response mode can therefore soften the effective *static* gravitational stiffness when it is allowed to relax.

For one normalized mode,

$$K_G^{\text{eff}} = K_G - \frac{|m|^2}{K_I}, \quad (516)$$

and positivity requires

$$|m|^2 < K_G K_I. \quad (517)$$

If

$$K_I \rightarrow 0^+ \quad (518)$$

while K_G remains finite and positive, stability requires

$$|m| = O(\sqrt{K_I}) \quad (519)$$

or faster.

This result must not be extended without qualification to arbitrary unbounded differential operators. In that case the Schur complement requires domain, self-adjointness, positivity, and closed-quadratic-form conditions.

Nor should the static positivity theorem be applied directly to a general retarded kernel. At finite frequency,

$$\Gamma_R(\omega, \mathbf{k})$$

is generally complex and dissipative. Its Schur complement modifies poles, widths, residues, and dispersion relations but does not satisfy a universal positive-softening ordering.

8.11 Gravitational versus internal comparison

A further structural distinction concerns the gravitational and internal comparison sectors.

The Lorentz connection

$$\omega$$

compares geometry-visible Lorentz data, whereas an internal comparison connection

$$A$$

acts on internal or multiplicity degrees of freedom.

A combined covariant derivative may possess the tensor-product structure

$$\nabla = \nabla_{\text{geom}} \otimes I + I \otimes \nabla_{\text{int}}, \quad (520)$$

without implying

$$A = \omega$$

or any inclusion relation between their gauge groups.

Thus

$$\text{coexistence of two connections} \not\Rightarrow \text{direct connection mixing}. \quad (521)$$

A quadratic mixing term requires an invariant intertwiner. For unbroken product symmetry

$$\text{Spin}^+(1, 3) \times G_{\text{int}},$$

a necessary representation-theoretic condition is

$$\text{Hom}_{\text{Spin}^+(1,3) \times G_{\text{int}}}(R_I, R_G) \neq 0. \quad (522)$$

If the gravitational mode is an internal singlet while the internal mode contains no singlet component, direct bilinear mixing is forbidden as long as the product symmetry remains unbroken.

This prevents a misleading identification of internal gauge symmetry with spacetime Lorentz geometry merely because both are described by connections.

8.12 Relation to the broader MS program

The broader MS quantum program can now be organized into distinct branches.

The quantum-soldering construction concerns the passage from operational geometry to canonical quantum deformation of soldering data.

The relational-multiplicity construction concerns quantum information carried by multiplicity spaces that are invisible to the collective geometry-visible representation, together with the possible emergence of internal comparison connections.

The present paper develops the complementary gravitational-response branch:

$$\begin{aligned} \text{spin-cover complex structure} &\longrightarrow \text{Lorentz bivector complex structure}, \\ \text{bivector rigidity} &\longrightarrow J_G = \pm \star, \\ \text{constitutive compatibility} &\longrightarrow C \propto \star, \\ \text{quantum source} &\longrightarrow \text{source-induced operator geometry}, \\ \text{response theory} &\longrightarrow \text{gravitational stiffness and noise}, \\ \text{infrared dynamics} &\longrightarrow \text{candidate Einstein response regime}. \end{aligned} \quad (523)$$

These branches address different questions and should not be collapsed into a single kinematical identification.

Taken together, they do not yet constitute a complete microscopic theory.

8.13 Open problems

Several decisive problems remain open.

First, the *common-spin-cover problem*: the present paper determines the Hodge complex structure once the quantum comparison carrier and the operational Lorentz geometry are assumed to use the same $\text{Spin}^+(1, 3)$ structure. The dynamical origin of that common realization remains to be derived.

Second, the *comparison-complex representation problem*: the quarter-turn J_ζ on the doubled constitutive response selects the pure Hodge graph, but a microscopic derivation must explain why the primitive comparison complex structure acts on the gravitational response variables through that particular exchange map.

Third, the *intrinsic quantum-geometry problem*: source transduction produces operator-valued geometric correlations, but these may be entirely inherited from quantum matter. A full quantum gravitational theory requires independent geometric observables, their operator algebra, vacuum correlations, and causal dynamics.

Fourth, the *field-theoretic soldering problem*: the local quantum-soldering construction must be extended to an infinite-dimensional field phase space with constraints, gauge reduction, renormalization, and a physical state space.

Fifth, the *gravitational-stiffness problem*: the response analysis provides an operational meaning for G through K_G^{-1} , but it does not predict the numerical infrared value of K_G .

Sixth, the *infrared-closure problem*: one must show dynamically that the long-distance response approaches a finite-stiffness pure Hodge channel while higher-derivative, nonlocal, and additional constitutive operators become irrelevant or controlled.

Seventh, the *intrinsic-noise problem*: the schematic separation

$$N_{GG}^{\text{full}} = N_{GG}^{\text{source}} + N_{GG}^{\text{intrinsic}} + N_{GG}^{\text{mixed}} \quad (524)$$

must be defined precisely within a microscopic operator algebra or a controlled effective open-system framework.

Eighth, the *comparison-sector mixing problem*: if a soft or massless internal comparison field emerges, its interaction with gravity must satisfy the representation-theoretic and stability conditions identified above. The present Schur-complement analysis describes only the quadratic static boundary of this problem.

Ninth, the *chirality and spin-statistics problem*: the spinorial construction identifies the two Weyl chiralities and their relation to the two Hodge eigenspaces, but it does not dynamically select one chirality. Likewise, projective spin parity does not by itself establish fermionic exchange statistics.

Tenth, the *phenomenological internal-structure problem*: the present comparison framework does not derive the Standard Model gauge group, its representations, hypercharges, generations, or anomaly cancellation. Numerical coincidences among low-dimensional multiplicity spaces are not sufficient to establish such a derivation.

8.14 Logical status of the results

The strongest conclusions of the present paper fall into three logical classes.

The first class consists of mathematical consequences within explicitly stated representation or response assumptions. These include: the Lorentz bivector commutant classification; the result

$$J_G = \pm \star$$

for Lorentz-equivariant real complex structures in four dimensions; the conditional bulk-dimensional rigidity of the bivector carrier; the helicity–Chern dimensional theorem under the rank-one line-bundle assumption; the exact constitutive-graph theorem; and the linear source-transduction identities once a deterministic response map and an operator-ordering prescription have been specified.

The second class consists of conditional structural identifications. These include the transfer of the spin-cover complex structure to the Hodge complex structure through a common spin cover; the selection of the pure Hodge constitutive direction through comparison-complex graph invariance; and the interpretation of source-induced torsional operators as operator-valued geometric response.

The third class consists of dynamical targets. These include selection of the relevant microscopic comparison carrier, emergence of the common Lorentz spin cover, derivation of the doubled response complex structure, preservation or attraction of the Hodge direction under coarse graining, finite infrared gravitational stiffness, intrinsic quantum gravitational degrees of freedom, and the dynamical relation between gravitational and internal comparison sectors.

Keeping these three levels distinct is necessary in order to avoid circularity.

8.15 Conceptual synthesis

The principal synthesis of the paper can be summarized as

$$\begin{aligned}
 &\text{quantum comparison} \longrightarrow \text{complex structure,} \\
 &\text{common Lorentz spin cover} \longrightarrow \text{Hodge complex structure,} \\
 &\text{bivector complex rigidity} \longrightarrow \text{conditional four-dimensional selection,} \\
 &\text{comparison–constitutive compatibility} \longrightarrow \text{pure-Hodge candidate response,} \\
 &\text{constitutive stiffness} \longrightarrow \text{gravitational compliance,} \\
 &\text{quantum sources} \longrightarrow \text{source-induced correlated geometry,} \\
 &\text{quantum response theory} \longrightarrow \text{causal susceptibility and noise,} \\
 &\text{infrared dynamics} \longrightarrow \text{candidate Einstein regime.}
 \end{aligned} \tag{525}$$

The significance of this chain lies as much in its separations as in its connections.

The scalar imaginary unit is not literally the Hodge star.

A projective spinorial comparison sector is not automatically a Weyl quantum field.

A Weyl carrier is not automatically a fermionic field with a specified spin–statistics relation.

A source-induced torsion operator is not automatically an independent quantum gravitational degree of freedom.

Quantum geometric noise is not automatically intrinsic gravitational noise.

A Planck-scale conjugacy parameter is not automatically a minimum length or minimum area.

A Lorentz-equivariant local constitutive plane is not automatically closed under renormalization.

A pure-Hodge infrared direction is not determined by an unnormalized response operator that simultaneously vanishes in magnitude.

A static Schur-complement softening theorem is not automatically a theorem about the full causal retarded kernel.

A response interpretation of Newton’s coupling is not a numerical prediction of Newton’s constant.

Within these boundaries, the present construction provides a coherent conditional gravitational extension of the MS comparison program.

Four-dimensional Lorentz geometry is singled out within the specified adjoint/bivector representation class as the one-time Lorentz dimension in which the gravitational generator carrier possesses the required invariant real complex structure.

Under the common-spin-cover compatibility assumption, that structure is the Hodge complex structure.

Under the additional comparison–constitutive graph-invariance assumption, the same Hodge direction becomes the distinguished local constitutive response direction.

Quantum matter sources are then transduced through the gravitational response kernel into operator-valued geometric correlations, while the magnitude of the constitutive response acquires the operational meaning of gravitational stiffness.

What remains is predominantly dynamical rather than representation theoretic: to construct a microscopic comparison theory whose quantum dynamics selects the required common structures and whose infrared response actually approaches a stable finite-stiffness Einstein regime.